

TRIPLE ERGODIC AVERAGES WITH DISTINCT INTEGER SLOPES

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ABSTRACT. For every invertible mixing probability-preserving transformation, triple ergodic averages of bounded measurable functions along any three pairwise distinct nonzero integer slopes converge almost everywhere to the product of their integrals. The slopes may be positive or negative.

1. INTRODUCTION

Let $(X, \mathcal{F}, \mu)$ be a probability space and let $T : X \rightarrow X$ be an invertible measure-preserving transformation with measurable inverse. Given bounded measurable functions f_1, f_2, f_3 and distinct nonzero integers $\lambda_1, \lambda_2, \lambda_3$, consider

$$(1.1) \quad A_N(x) = \frac{1}{N} \sum_{n=1}^N \prod_{j=1}^3 f_j(T^{\lambda_j n} x).$$

The issue is convergence at almost every point, rather than only in norm. We assume that T is mixing, in the sense that

$$(1.2) \quad \mu(A \cap T^{-n} B) \longrightarrow \mu(A)\mu(B) \quad \text{as } |n| \longrightarrow \infty \quad (A, B \in \mathcal{F}).$$

Theorem 1.1. *For every system satisfying (1.2), every triple of pairwise distinct nonzero integers $\lambda_1, \lambda_2, \lambda_3$, and every bounded measurable $f_1, f_2, f_3 : X \rightarrow \mathbb{C}$, the averages in (1.1) satisfy*

$$A_N(x) \longrightarrow \prod_{j=1}^3 \int_X f_j d\mu \quad \text{for almost every } x,$$

as N tends to infinity through all positive integers. The exceptional null set may depend on the slopes and the functions.

The probability space need not be standard or countably generated. Both positive and negative slopes are allowed. The proof depends on the analytic results of the companion manuscripts [H, V]; their role and the extension required here are described below.

1.1. Context and predecessors. Birkhoff's pointwise ergodic theorem [Bir] treats one function. Products along several iterates arise naturally in Furstenberg's ergodic proof of the multiple recurrence theorem and its connection with arithmetic progressions [Fur]. Multiple recurrence, norm convergence, and pointwise convergence are different questions. In particular, convergence in L^2 does not by itself establish convergence at almost every point along the full sequence.

For weakly mixing transformations, the product-of-integrals limit in L^2 is classical. Bergelson's weakly mixing PET theorem [Ber, Theorem 1.2 and p. 340] includes distinct nonzero integer slopes of either sign. For general systems, Host and Kra [HK] and Ziegler [Zie] developed the norm-convergence theory using characteristic factors with nilpotent structure. These results explain the limiting structure without resolving the corresponding pointwise question.

Bourgain’s double recurrence theorem [Bou] establishes almost-sure convergence for two bounded functions evaluated at two powers of one transformation. Pointwise theorems for longer products also have substantial precedents under additional assumptions on the system. For example, Derrien and Lesigne [DL, Theorem 4.2] prove polynomial pointwise theorems for K -systems and exact endomorphisms; their invertible case includes the linear configurations considered here. Other structural hypotheses yielding pointwise convergence for all finite progression lengths include singular spectrum on the Pinsker factor of a weakly mixing system, by Assani [Ass], and the pairwise-independent-joining property for weakly mixing systems, by Gutman, Huang, Shao, and Ye [GHSY]. The significance of Theorem 1.1 is that its dynamical assumption is mixing alone, while the three slopes are arbitrary distinct nonzero integers.

The harmonic-analytic method separates convergence from identification of the limit. A uniform oscillation estimate gives pointwise convergence; mixing identifies that limit by an L^2 argument. Oscillation methods in ergodic theory are developed in [Bou, JKRW], and the passage from translation estimates to orbit estimates follows the transference principle of Calderón [Cal]. We give that passage directly, so it imposes no additional structure on the probability space.

1.2. Analytic input and the coefficient extension. The manuscript [H] proves an L^3 bound for the trilinear Hilbert transform through local four-linear forms. Its proof includes chart detection, localization by quadratic data, comparison on short depth blocks, and a separated-frequency counting estimate. The quantitative inverse theorem of Leng, Sah, and Sawhney [LSS] enters its chart detection argument. Arithmetic regularity and inverse-theorem methods also occur in Tao’s sublogarithmic cancellation theorem for truncated multilinear Hilbert transforms [Tao]; that cancellation theorem is distinct from the uniform local estimate used here.

The companion [V] adapts the local estimates to inputs that vary with depth and deduces pointwise convergence for the slopes 1, 2, 3. The local forms in both companions use the four coordinates $x, x + t, x + 2t, x + 3t$. A power of T and a permutation of the functions do not turn an arbitrary triple into this configuration. Thus their stated theorems cannot simply be invoked for the slopes in Theorem 1.1.

The new work is the extension of their local proofs to $x + b_j t$, $0 \leq j \leq 3$, for fixed distinct integers with $b_0 = 0$. Three features are particularly useful beyond this application. First, an integer left inverse of the *whole* primitive configuration replaces a unimodular pair of coordinates; this removes extra lattice or torus configurations in discretization. Second, a Vandermonde relation provides the quadratic cancellation and the Fourier randomization for every set of four distinct slopes. Third, a height base divisible by all pair differences preserves the divisibility required in the separated-frequency argument. The last step must preserve a bound polynomial in the dimension: an exponential loss there would not fit the proof’s parameter choices.

These are extensions of the proofs in [H, V], not new proofs of their coefficient-independent projection and row estimates. We state the local conclusion completely, prove the coefficient-sensitive identities and estimates, and specify the results imported from the companions at their points of use. In particular, the technical part is a proof by reference to those manuscripts, whereas the ergodic deduction is presented in full.

1.3. Proof structure. Section 2 defines the admissible kernels and states the local estimate for inputs with controlled variation on blocks of depths. Section 3 proves Theorem 1.1 from that estimate: smooth scale differences are realized by cancellative kernels, then transferred to integer sequences and finite orbits. The resulting oscillation bound gives convergence for smooth profiles; moment-corrected approximations to interval indicators recover all integer lengths.

The remaining sections prove the local estimate. Section 4 establishes the coefficient modifications in detection and localization. Section 5 proves the short-block, height, and counting modifications. Section 6 closes the parameter argument, first for fixed inputs and then for depth-dependent inputs. All analytic constants may depend on the fixed slopes; no bound is required uniformly over triples of integers.

2. THE LOCAL ESTIMATE AND THE COEFFICIENT ALGEBRA

The analytic estimate concerns four inputs on the real line. Its constant may depend on the slopes, but must be independent of the number of scales and of the number of changes of the inputs. This uniformity will give the oscillation estimate in Section 3.

2.1. Normalization and local forms. Set

$$g = \gcd(|\lambda_1|, |\lambda_2|, |\lambda_3|), \quad S = T^g, \quad b_0 = 0, \quad b_j = \lambda_j/g \quad (1 \leq j \leq 3).$$

Then S is invertible, probability preserving, and mixing, and $T^{\lambda_j n} = S^{b_j n}$. Henceforth the four distinct integers b_0, b_1, b_2, b_3 are fixed, and $\gcd(|b_1|, |b_2|, |b_3|) = 1$. We make no assertion of uniformity as these integers vary.

Dyadic intervals are half-open. The depth of a descendant I of a dyadic interval J is $s(I) = \log_2(|J|/|I|)$. A finite rooted tree contains every ancestor between any of its intervals and its root. The dyadic lattice may be translated.

Definition 2.1 (Local kernels). Fix $\delta_0 > 0$ and $C_0 \geq 1$. For an interval I of length r , a kernel $w_I \in C_c^\infty(\mathbb{R}^2; \mathbb{C})$ is admissible if

- (1) $\text{dist}(x + b_j t, \partial I) \geq \delta_0 r$ and $x + b_j t \in I$ for every $(x, t) \in \text{supp } w_I$ and $0 \leq j \leq 3$;
- (2) for every $v \in \mathbb{R}$ and $0 \leq j \leq 3$,

$$(2.1) \quad \int_{\mathbb{R}} w_I(v - b_j t, t) dt = 0;$$

- (3) for nonnegative integers a, d and $n = a + d$,

$$(2.2) \quad \|\partial_x^a \partial_t^d w_I\|_\infty \leq r^{-n} (C_0(n+2))^{C_0(n+2)}.$$

For scalar measurable inputs define

$$(2.3) \quad H_I(z_0, z_1, z_2, z_3) = r^{-2} \int_{\mathbb{R}^2} w_I(x, t) \prod_{j=0}^3 z_j(x + b_j t) dx dt,$$

whenever the integral is absolutely convergent.

For a nonempty integer block $B = \{u, \dots, v\}$ and functions z_s on a common domain, put

$$\mathcal{V}_B(z) = \max_{u \leq s \leq v} |z_s| + \sum_{s=u}^{v-1} |z_{s+1} - z_s|.$$

This is a pointwise bound on both size and variation within the block; no variation across block boundaries is included.

Theorem 2.2 (Local estimate with variation in depth). *Fix the primitive distinct integers $b_0 = 0, b_1, b_2, b_3$ and the parameters δ_0, C_0 . Let $\mathcal{T}$ be a finite dyadic tree with root J and largest depth d , and let $\mathcal{I} \subseteq \mathcal{T}$. Assign an admissible kernel to every $I \in \mathcal{I}$. For each $0 \leq j \leq 3$, let $z_{j,s} : J \rightarrow \mathbb{C}$, $0 \leq s \leq d$, be measurable, and partition $\{0, \dots, d\}$ into $W_j \geq 1$ deterministic nonempty consecutive blocks. Suppose that*

$$\mathcal{V}_B(z_j) \leq W_j^{-1/2} \quad \text{almost everywhere on } J$$

for each block in the j th partition. Extend every input by zero outside J . Then

$$(2.4) \quad \sum_{I \in \mathcal{I}} |I| |H_I(z_{0,s(I)}, z_{1,s(I)}, z_{2,s(I)}, z_{3,s(I)})| \leq C_{b,\delta_0,C_0} |J|.$$

The constant is independent of the lattice, tree, depth range, subcollection, kernels, and the four partitions and their sizes.

The schedules are prescribed at every depth, even depths with no summation intervals. This is exactly the hypothesis of [V, Theorem 3.2], except for the coefficients in the local forms. Sections 4–6 prove the theorem by extending that proof and its analytic inputs from [H]. Before turning to those technical arguments, Section 3 proves the implication from Theorem 2.2 to the ergodic theorem.

2.2. Integer reconstruction and quadratic cancellation. Two features of the coefficients recur in both the analytic proof and transference. Pairs of coordinates determine (x, t) over $\mathbb{R}$, but need not determine it over $\mathbb{Z}$. All four coordinates together do determine the integral step. Quadratic cancellation requires a separate identity, furnished by Vandermonde interpolation.

Lemma 2.3 (Coefficient identities). *There exist integers e_0, e_1, e_2, e_3 and nonzero integers c_0, c_1, c_2, c_3 such that*

$$(2.5) \quad \sum_{j=0}^3 e_j = 0, \quad \sum_{j=0}^3 e_j b_j = 1,$$

and

$$(2.6) \quad \sum_{j=0}^3 c_j b_j^a = 0 \quad (a = 0, 1, 2).$$

The nullspace over $\mathbb{R}$ of the two rows $(1)_j, (b_j)_j$ is spanned by $(c_j)_j$ and $(b_j c_j)_j$. Consequently, for $y_j = x + b_j t$,

$$x = y_0, \quad t = \sum_j e_j y_j, \quad \sum_j c_j y_j^2 = 0.$$

For any distinct i, j , the pair map $(x, t) \mapsto (x + b_i t, x + b_j t)$ is injective on $\mathbb{Z}^2$ and pushes normalized Haar measure on $(\mathbb{R}/L\mathbb{Z})^2$ to normalized Haar measure, for every $L > 0$.

Proof. Bezout's identity gives integers e_1, e_2, e_3 with $\sum_{j=1}^3 e_j b_j = 1$; set $e_0 = -\sum_{j=1}^3 e_j$. Let $P(z) = \prod_{j=0}^3 (z - b_j)$. Lagrange interpolation of z^a , $0 \leq a \leq 2$, followed by comparison of the coefficient of z^3 , gives $\sum_j b_j^a / P'(b_j) = 0$. Multiply by a common positive integer multiple of the denominators to obtain (2.6) with every $c_j \neq 0$. The two displayed null vectors are independent because the b_j are distinct, and the nullspace has dimension two. The identities for y_j follow by expansion. Finally the pair matrix has determinant $b_j - b_i \neq 0$, so it is injective on the lattice and surjective on the real torus. A surjective continuous homomorphism of compact groups pushes normalized Haar measure to normalized Haar measure. $\square$

Remark 2.4. Primitivity does not mean that some pair is unimodular. For example, the slopes $(b_0, b_1, b_2, b_3) = (0, 6, 10, 15)$ are primitive, but no pair difference is 1 in absolute value. This is why the integer inverse in (2.5), rather than division by a pair determinant, is used in the lattice and torus arguments below.

2.3. The fixed-input constant. The proof of Theorem 2.2 first establishes a fixed-input estimate. For $2 < q < 3$, let $M_q f = (M(|f|^q))^{1/q}$, where M is the uncentered Hardy–Littlewood maximal operator. Let $C_N(q)$ be the least constant for which

$$(2.7) \quad \sum_{I \in \mathcal{A}} |I| |H_I(f_0, f_1, f_2, f_3)| \leq C_N(q) \int_{\mathbb{R}} \prod_{j=0}^3 M_q f_j(x) dx$$

for all finite collections $\mathcal{A}$ of intervals in one translated dyadic lattice, whose lengths belong to at most N consecutive scales, admissible kernels, and compactly supported scalar L^q inputs. This is the definition in [H, Section 2], with our coefficients. Pair-coordinate estimates imply $C_N(q) \lesssim_b N$, so this constant is finite before any absorption argument. The analytic proof will show $\sup_N C_N(q) < \infty$ for a suitable fixed $q > 2$, and then use that bound in the scheduled-input argument. This order avoids using Theorem 2.2 to prove its own analytic input.

3. FROM THE LOCAL ESTIMATE TO POINTWISE CONVERGENCE

We now deduce Theorem 1.1 from Theorem 2.2. This section is independent of the internal notation of the analytic proof. It follows the transference scheme of [V, Sections 2–3], including its use of signed smooth profiles. We give all the details, both to handle arbitrary integer slopes and to separate the two roles of the hypotheses: the local estimate gives almost everywhere convergence of smooth averages on every invertible probability preserving system; mixing identifies their limit.

Throughout, $b_0 = 0, b_1, b_2, b_3$ are the fixed primitive slopes, S is an invertible measure-preserving transformation, and e_j are the integers in (2.5). All constants in this section may depend on the slopes.

3.1. Smooth profiles and cancellation kernels.

Definition 3.1 (Averaging profile). An averaging profile is a real $\varphi \in C_c^\infty(\mathbb{R})$ with

$$(3.1) \quad \int_{\mathbb{R}} \varphi = 1, \quad \int_{\mathbb{R}} t^j \varphi(t) dt = 0 \quad (1 \leq j \leq 3),$$

and, for some $G \geq 1$,

$$(3.2) \quad \|\varphi^{(n)}\|_\infty \leq (G(n+2))^{G(n+2)} \quad (n \geq 0).$$

Write $\varphi_a(t) = a^{-1} \varphi(t/a)$ for $a > 0$. Profiles are necessarily signed: a nonnegative smooth function of mass one cannot have zero second moment.

The derivative class in (3.2) is closed under fixed translations, dilations, differentiation, and finite linear combinations. It contains a nonnegative bump ρ of mass one supported in $[-1, 1]$. For completeness, normalize

$$\rho_0(t) = \begin{cases} \exp(-1/(1-t^2)), & |t| < 1, \\ 0, & |t| \geq 1. \end{cases}$$

For $|t| < 1$, put $d = 1 - t^2$. On a complex disk of radius cd about t , with sufficiently small absolute $c > 0$, the real part of $(1 - z^2)^{-1}$ is at least c'/d . Cauchy's estimate gives

$$|\rho_0^{(n)}(t)| \leq n!(cd)^{-n} e^{-c'/d}.$$

For each n this tends to zero at the endpoints, and its maximum over $0 < d \leq 1$ is at most $C^{n+1} n!(n+1)^n$. Thus the zero extension is smooth and satisfies (3.2) after enlarging G . Lemma 3.9 will construct averaging profiles arbitrarily close to interval indicators.

Fix a profile φ , and put

$$(3.3) \quad \psi = \varphi_2 - \varphi, \quad \Delta_k(t) = \varphi_{2^{k+1}}(t) - \varphi_{2^k}(t) = 2^{-k}\psi(2^{-k}t).$$

Lemma 3.2 (Realizing a scale difference). *There exist a dyadic integer $D \geq 1$, a constant $C_0 \geq 1$, and a real function $\mathcal{W} \in C_c^\infty(\mathbb{R}^2)$, depending only on φ and the slopes, with the following property. For an interval I of length $r = D2^k$ and midpoint m_I , the kernel*

$$w_I(x, t) = \mathcal{W}\left(\frac{x - m_I}{r}, \frac{t}{r}\right)$$

is admissible for Theorem 2.2, with interior margin $\delta_0 = 1/8$ and derivative constant C_0 . Moreover,

$$(3.4) \quad \frac{1}{r} \int_{\mathbb{R}} \mathcal{W}(z, t/r) dz = \Delta_k(t) \quad (t \in \mathbb{R}).$$

Proof. The moments of ψ through order three vanish: its j th moment is $(2^j - 1) \int t^j \varphi(t) dt$. Choose $B \geq 1$ with $\text{supp } \psi \subseteq [-B, B]$, and define

$$g(t) = \frac{1}{6} \int_{-\infty}^t (t - v)^3 \psi(v) dv.$$

For $t > B$, expansion of the cubic and the four zero moments show that $g(t) = 0$; it also vanishes for $t < -B$. Hence $g \in C_c^\infty(\mathbb{R})$, $g^{(4)} = \psi$, and g satisfies (3.2) with a possibly larger constant. Indeed, its first four derivative bounds follow from integration over $[-B, B]$, and its subsequent derivatives are derivatives of ψ .

Take $u(z) = 8\rho(8z)$, with the bump constructed above. Choose a dyadic integer $D > 4B \max_j |b_j|$, and set

$$F(z, \sigma) = u(z)D^{-3}g(D\sigma), \quad L_j = \partial_\sigma - b_j\partial_z, \quad \mathcal{W} = \prod_{j=0}^3 L_j F.$$

The constant-coefficient operators commute. On the support of $\mathcal{W}$,

$$|z + b_j\sigma| \leq \frac{1}{8} + \frac{|b_j|B}{D} < \frac{3}{8}.$$

After rescaling, every $x + b_j t$ therefore lies in I , at distance greater than $r/8$ from its endpoints. Put $G_j = \prod_{i \neq j} L_i F$. The chain rule gives

$$w_I(v - b_j t, t) = r \frac{d}{dt} G_j \left(\frac{v - b_j t - m_I}{r}, \frac{t}{r} \right).$$

The function differentiated is compactly supported in t , so its integral is zero. This proves all four line cancellations. The product rule and (3.2) give the required derivative bounds, with fixed D, b_j absorbed into C_0 .

Finally, integration in z kills every term of $\prod_j L_j F$ containing a z -derivative. Since $\int u = 1$,

$$\int_{\mathbb{R}} \mathcal{W}(z, \sigma) dz = Dg^{(4)}(D\sigma) = D\psi(D\sigma).$$

Division by $r = D2^k$ yields (3.4). $\square$

We use the block size and variation $\mathcal{V}_B$ from Section 2. Block endpoints are always deterministic; the input values may depend on the spatial variable.

Lemma 3.3 (A real-line test). *Let $0 \leq k_- \leq k_+$, and partition $\{k_-, \dots, k_+\}$ into w nonempty consecutive blocks. Suppose that bounded Borel functions h_1, h_2, h_3 and $h_{0,k}$ are supported in a common interval of length $L > 0$, that $|h_j| \leq 1$ for $1 \leq j \leq 3$, and that $\mathcal{V}_{B_0}(h_0) \leq w^{-1/2}$ on every block B_0 . Then*

$$(3.5) \quad \left| \sum_{k=k_-}^{k_+} \int_{\mathbb{R}^2} h_{0,k}(x) \prod_{j=1}^3 h_j(x + b_j t) \Delta_k(t) dx dt \right| \leq C_{\varphi,b}(L + D2^{k_+}).$$

The constant is independent of the scale range, partition, and inputs.

Proof. Put $r_k = D2^k$ and $R = r_{k_+}$. For $0 \leq a < R$, use the translated dyadic lattice

$$\mathcal{D}_a = \{[a + m2^q, a + (m+1)2^q) : m, q \in \mathbb{Z}\}.$$

At scale r_k , use the kernels from Lemma 3.2. Only roots of length R meeting the common support interval can contribute: the zeroth coordinate of a supported kernel belongs to its interval. These roots have total length at most $L + 2R$.

Restrict all inputs to each such root. This changes no local form, since all four coordinates on its kernel support belong to the summation interval. In the root tree, depth is $s = k_+ - k$. Reversing the scale order preserves the block size and variation. Thus slot zero satisfies the schedule hypothesis of Theorem 2.2, and the three fixed slots satisfy it with one block each. Summing that theorem over the roots gives

$$(3.6) \quad \left| \sum_{k=k_-}^{k_+} \sum_{\substack{I \in \mathcal{D}_a \\ |I|=r_k}} r_k H_I(h_{0,k}, h_1, h_2, h_3) \right| \leq C_{b,1/8,C_0}(L + 2R).$$

For $r = r_k$, the following sum is r -periodic in a . Since r divides R , unfolding one period yields

$$\begin{aligned} \frac{1}{R} \int_0^R \sum_{\substack{I \in \mathcal{D}_a \\ |I|=r}} r^{-1} w_I(x, t) da &= \frac{1}{r} \int_0^r \sum_{m \in \mathbb{Z}} r^{-1} \mathcal{W}\left(\frac{x - a - (m+1/2)r}{r}, \frac{t}{r}\right) da \\ &= \frac{1}{r} \int_{\mathbb{R}} \mathcal{W}(z, t/r) dz = \Delta_k(t). \end{aligned}$$

As rH_I is the integral against $r^{-1}w_I$, averaging (3.6) in a gives (3.5). All exchanges of integrals and sums involve finitely many scales, bounded supports, and uniformly finitely many relevant intervals. $\square$

3.2. Integer sequences and finite orbit segments. The next step is where primitivity is useful. We do not assume that any pair of slopes has difference one.

Lemma 3.4 (An integer-sequence test). *Under the scale and block hypotheses of Lemma 3.3, let h_1, h_2, h_3 and $h_{0,k}$ instead be sequences on $\mathbb{Z}$, supported in a common integer interval of cardinality $L \geq 1$. Assume the same pointwise size and block variation bounds. Then*

$$(3.7) \quad \left| \sum_{k=k_-}^{k_+} \sum_{m,n \in \mathbb{Z}} h_{0,k}(m) \prod_{j=1}^3 h_j(m + b_j n) \Delta_k(n) \right| \leq C'_{\varphi,b}(L + D2^{k_+}).$$

Proof. Write $E_b = \sum_j |e_j|$, and fix $0 < \rho < 1/4$ so small that

$$(3.8) \quad \rho(2 + \max_j |b_j| E_b) < 1.$$

Plant every sequence value on the half-open interval $[m - \rho, m + \rho)$, and set it equal to zero elsewhere. The resulting Borel functions have the same size and variation bounds, and their common real support lies in an interval of length at most $L - 1 + 2\rho \leq L$.

Suppose a configuration lies in four planted intervals, with integer centers p_j , and write $\delta_j = x + b_j t - p_j$, so $|\delta_j| < \rho$ away from null boundaries. Set

$$m = p_0, \quad n = \sum_j e_j p_j, \quad \xi = x - m, \quad \eta = t - n.$$

By (2.5), $\eta = \sum_j e_j \delta_j$. Consequently

$$|p_j - (m + b_j n)| \leq |\delta_j| + |\xi| + |b_j \eta| < \rho(2 + |b_j| E_b) < 1.$$

The expression inside the absolute value is an integer, so $p_j = m + b_j n$ for all four slots. Conversely, a configuration lies in the intervals with these centers precisely when

$$(\xi, \eta) \in P_\rho := \{(\xi, \eta) \in \mathbb{R}^2 : |\xi + b_j \eta| < \rho \ (0 \leq j \leq 3)\},$$

up to boundaries. This polygon has positive finite area A_ρ : it contains a neighborhood of zero and is bounded because two distinct coordinate forms are linearly independent. Distinct integer pairs (m, n) give disjoint regions up to boundaries, by uniqueness of the centers and (2.5). On P_ρ , $|\eta| \leq \rho E_b$.

It follows that the real-line integral at scale k is exactly

$$I_k = \sum_{m, n \in \mathbb{Z}} h_{0,k}(m) \prod_{j=1}^3 h_j(m + b_j n) \int_{P_\rho} \Delta_k(n + \eta) d\xi d\eta.$$

Let D_k be the discrete sum at scale k in (3.7). The derivative bound $\|\Delta'_k\|_\infty = 2^{-2k} \|\psi'\|_\infty$ gives

$$|\Delta_k(n + \eta) - \Delta_k(n)| \leq \rho E_b \|\psi'\|_\infty 2^{-2k}.$$

If this difference is nonzero, then $|n| \leq B2^k + \rho E_b$. Since $k \geq 0$, there are $O_{\varphi,b}(2^k)$ such integers n , and at most L contributing integers m . Every coefficient product has absolute value at most one. Hence

$$|I_k - A_\rho D_k| \leq A_\rho C_{\varphi,b} L 2^{-k}.$$

Sum over k , use $\sum_{k \geq 0} 2^{-k} = 2$, and apply Lemma 3.3 to the planted inputs. Division by $A_\rho > 0$ proves (3.7). $\square$

For measurable functions $|f_j| \leq 1$ on the probability space, put

$$(3.9) \quad F_n(x) = \prod_{j=1}^3 f_j(S^{b_j n} x), \quad V_k^\varphi(x) = \sum_{n \in \mathbb{Z}} \varphi_{2^k}(n) F_n(x) \quad (k \geq 0).$$

All sums are finite. If $\text{supp } \varphi \subseteq [-B_\varphi, B_\varphi]$, with $B_\varphi \geq 1$, then

$$(3.10) \quad |V_k^\varphi(x)| \leq (2B_\varphi + 1) \|\varphi\|_\infty.$$

Proposition 3.5 (Finite orbit oscillation). *For every deterministic list $0 \leq k_1 < \dots < k_{w+1}$,*

$$(3.11) \quad \sum_{i=1}^w \int_X \max_{k_i \leq \ell \leq k_{i+1}} |V_\ell^\varphi - V_{k_i}^\varphi| d\mu \leq C''_{\varphi,b} \sqrt{w}.$$

This holds on every invertible probability preserving system, with a constant independent of that system, the functions, and the endpoints.

Proof. First take sequences h_j , bounded by one and supported in an integer interval $\mathcal{K}$ of cardinality L , and set

$$V_k(m) = \sum_n \varphi_{2^k}(n) \prod_{j=1}^3 h_j(m + b_j n).$$

For m in a subset $\mathcal{E} \subseteq \mathcal{K}$, let $\ell_i(m)$ be the smallest maximizing index of $|V_\ell(m) - V_{k_i}(m)|$ in the finite range $k_i \leq \ell \leq k_{i+1}$. Write $d_i(m) = V_{\ell_i(m)}(m) - V_{k_i}(m)$, and take $\alpha_i(m) = \overline{d_i(m)}/|d_i(m)|$ when $d_i(m) \neq 0$, and zero otherwise. In the block $k_i \leq k < k_{i+1}$, set

$$h_{0,k}(m) = \begin{cases} \alpha_i(m)/(4\sqrt{w}), & m \in \mathcal{E}, k < \ell_i(m), \\ 0, & \text{otherwise.} \end{cases}$$

On each block this sequence has size at most $(4\sqrt{w})^{-1}$ and at most one jump of that size. Its block size and variation is therefore at most $w^{-1/2}$. The block endpoints remain deterministic, although the prefix and phase depend on m .

The identity $\Delta_k = \varphi_{2^{k+1}} - \varphi_{2^k}$ telescopes, so Lemma 3.4 gives

$$(3.12) \quad \sum_{m \in \mathcal{E}} \sum_{i=1}^w \max_{k_i \leq \ell \leq k_{i+1}} |V_\ell(m) - V_{k_i}(m)| \leq 4C'_{\varphi,b} \sqrt{w} (L + D2^{k_{w+1}}).$$

Indeed, the sum tested in that lemma is exactly $(4\sqrt{w})^{-1}$ times the nonnegative left side, since $\alpha_i(m)d_i(m) = |d_i(m)|$.

We now use finite orbit segments, the elementary form of Calderón's transference principle [Cal]. Fix the endpoints, and choose an integer

$$Q \geq B_\varphi \max_j |b_j| 2^{k_{w+1}}.$$

For each $M \geq 1$ and $x \in X$, use the sequences

$$h_{j,x}(r) = \begin{cases} f_j(S^r x), & 1 - Q \leq r \leq M + Q, \\ 0, & \text{otherwise.} \end{cases}$$

Their common support has cardinality $M + 2Q$. For $1 \leq m \leq M$ and all scales in the endpoint list, no term of the average is cut off, so $V_k(m) = V_k^\varphi(S^m x)$. Apply (3.12) with $\mathcal{E} = \{1, \dots, M\}$, integrate over x , and use invariance under S^m . This gives

$$M \sum_{i=1}^w \int_X \max_{k_i \leq \ell \leq k_{i+1}} |V_\ell^\varphi - V_{k_i}^\varphi| d\mu \leq 4C'_{\varphi,b} \sqrt{w} (M + 2Q + D2^{k_{w+1}}).$$

The maxima are measurable finite maxima. Divide by M and let $M \rightarrow \infty$, keeping the endpoints and hence Q fixed. This proves (3.11). $\square$

Lemma 3.6 (Oscillation implies convergence). *Let $(Z_k)_{k \geq 0}$ be a uniformly bounded sequence of measurable complex functions on a probability space. Suppose that, for every deterministic $0 \leq k_1 < \dots < k_{w+1}$,*

$$\sum_{i=1}^w \int_X \max_{k_i \leq \ell \leq k_{i+1}} |Z_\ell - Z_{k_i}| d\mu \leq C\sqrt{w}.$$

Then Z_k converges on a measurable conull set.

Proof. We include the standard finite-block argument; compare the oscillation methods in [Bou, JKRW]. The measurable function

$$\omega(x) = \lim_{K \rightarrow \infty} \sup_{p, q \geq K} |Z_p(x) - Z_q(x)|$$

vanishes exactly where the sequence is Cauchy. If its positive set has positive measure, there are $\delta > 0$ and $E = \{\omega > 2\delta\}$ with $\mu(E) = d > 0$. For every K ,

$$\sup_{\ell \geq K} |Z_\ell - Z_K| \geq \frac{1}{2}\omega > \delta \quad \text{on } E.$$

By monotone convergence of the finite maxima, a deterministic $L > K$ satisfies $\int \max_{K \leq \ell \leq L} |Z_\ell - Z_K| d\mu \geq \delta d/2$. Starting with $k_1 = 0$, choose successive endpoints this way. The assumed estimate would give $w\delta d/2 \leq C\sqrt{w}$ for every w , a contradiction. Thus $\omega = 0$ almost everywhere, and completeness of $\mathbb{C}$ proves the claim. $\square$

Proposition 3.5, (3.10), and Lemma 3.6 already give almost everywhere convergence of V_k^φ . Mixing has not yet been used.

3.3. Identifying the limit by mixing. We include the norm argument to make explicit that it accommodates arbitrary distinct nonzero slopes and both signs of time. The set-mixing assumption on S implies, by uniform approximation by simple functions, that for bounded measurable g, h ,

$$(3.13) \quad \int_X g(h \circ S^m) d\mu \longrightarrow \left(\int_X g d\mu \right) \left(\int_X h d\mu \right) \quad (|m| \rightarrow \infty).$$

Lemma 3.7 (The Hilbert space van der Corput criterion). *Let $(u_n)_{n \geq 1}$ be bounded vectors in a complex Hilbert space, with inner product linear in the first variable. If the limits*

$$\gamma_h = \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=1}^{N-h} \langle u_{n+h}, u_n \rangle$$

exist for each $h \geq 1$ and tend to zero as $h \rightarrow \infty$, then $N^{-1} \sum_{n=1}^N u_n \rightarrow 0$ in norm.

Proof. We record the finite estimate underlying the criterion; compare [HK, Lemma D.1]. Extend u_n by zero outside $\{1, \dots, N\}$. The identity

$$H \sum_{n=1}^N u_n = \sum_{m=1}^{N+H-1} \sum_{j=0}^{H-1} u_{m-j}$$

and Cauchy's inequality give

$$\begin{aligned} \left\| \frac{1}{N} \sum_{n=1}^N u_n \right\|^2 &\leq \frac{N+H-1}{N^2 H^2} \left(H \sum_{n=1}^N \|u_n\|^2 \right. \\ &\quad \left. + 2 \sum_{h=1}^{H-1} (H-h) \operatorname{Re} \sum_{n=1}^{N-h} \langle u_{n+h}, u_n \rangle \right). \end{aligned}$$

If $\|u_n\| \leq M$, first let $N \rightarrow \infty$ with H fixed, obtaining

$$\limsup_N \left\| \frac{1}{N} \sum_{n=1}^N u_n \right\|^2 \leq \frac{M^2}{H} + \frac{2}{H} \sum_{h=1}^{H-1} |\gamma_h|.$$

The right side tends to zero with H , since the Cesàro means of $(|\gamma_h|)$ tend to zero. $\square$

Lemma 3.8 (Norm convergence at distinct slopes). *If S is mixing, $a_1, \dots, a_r$ are distinct nonzero integers, and $g_1, \dots, g_r$ are bounded measurable functions, then*

$$(3.14) \quad \frac{1}{N} \sum_{n=1}^N \prod_{j=1}^r g_j \circ S^{a_j n} \longrightarrow \prod_{j=1}^r \int_X g_j d\mu \quad \text{in } L^2(\mu).$$

Consequently, for F_n in (3.9) and $p = \prod_{j=1}^3 \int_X f_j d\mu$, every $\chi \in C_c^1(\mathbb{R})$ of integral one satisfies

$$(3.15) \quad \sum_{n \in \mathbb{Z}} a^{-1} \chi(n/a) F_n \longrightarrow p \quad \text{in } L^2(\mu), \quad a \rightarrow \infty.$$

Proof. Write $U^m f = f \circ S^m$. These operators are unitary on $L^2(\mu)$, with inner product $\langle f, g \rangle = \int f \bar{g} d\mu$. We induct on r , starting with the empty product $r = 0$. Suppose first that one of the g_j has mean zero. Put

$$u_n = \prod_{j=1}^r U^{a_j n} g_j, \quad q_{j,h} = (U^{a_j h} g_j) \overline{g_j}.$$

By invariance,

$$\langle u_{n+h}, u_n \rangle = \int_X q_{1,h} \prod_{j=2}^r U^{(a_j - a_1)n} q_{j,h} d\mu.$$

For fixed h , induction applies to the distinct nonzero slopes $a_j - a_1$, $2 \leq j \leq r$. The average of this correlation therefore tends to

$$\gamma_h = \prod_{j=1}^r \int_X q_{j,h} d\mu.$$

Here removing the final h terms and dividing by N instead of $N - h$ does not change the limit. By (3.13), the j th factor tends to $|\int g_j d\mu|^2$ as $h \rightarrow \infty$. At least one limit is zero, so $\gamma_h \rightarrow 0$. Lemma 3.7 proves (3.14) in the centered case. For general inputs, write each g_j as its mean plus a mean-zero function and expand. The constant term is the asserted limit, and every other term contains a centered factor, so the centered case applies. This completes the induction. Only fixed h is used in the induction; no estimate uniform in h is needed.

Apply (3.14) to b_j and $-b_j$. The partial sums

$$R_N^\pm = \sum_{n=1}^N (F_{\pm n} - p) \quad \text{satisfy} \quad \|R_N^\pm\|_2 = o(N).$$

Choose $B \geq 1$ containing the support of χ , and put $M_a = \lfloor Ba \rfloor + 1$, $w_n^\pm = a^{-1} \chi(\pm n/a)$. Then $w_{M_a}^\pm = 0$, and summation by parts gives

$$\sum_{n=1}^{M_a} w_n^\pm (F_{\pm n} - p) = \sum_{n=1}^{M_a-1} (w_n^\pm - w_{n+1}^\pm) R_n^\pm.$$

Its norm is at most $a^{-2} \|\chi'\|_\infty \sum_{n < M_a} \|R_n^\pm\|_2 = o(1)$. To see the last assertion, split at a fixed N_0 , and use $\|R_n^\pm\|_2 \leq \epsilon n$ beyond N_0 ; the tail is $O_B(\epsilon)$, and the finite initial part tends to zero. The $n = 0$ term is $O(a^{-1})$ in norm, while the Riemann sum $a^{-1} \sum_n \chi(n/a)$ tends to one. Adding the two signs and the zero term proves (3.15). $\square$

For each averaging profile, Lemma 3.8 gives $V_k^\varphi \rightarrow p$ in L^2 . Its almost everywhere limit is therefore p . More explicitly, if E_φ is the measurable conull convergence set, Fatou's lemma gives

$$\int_{E_\varphi} |\lim_k V_k^\varphi - p|^2 d\mu \leq \liminf_k \|V_k^\varphi - p\|_2^2 = 0.$$

3.4. Removing the smooth weights. The moment restrictions need not obstruct approximation in L^1 . The correction below, from [V, Lemma 2.8], moves a small amount of signed mass to a large scale. Uniform control of the supports or derivative constants is neither asserted nor needed.

Lemma 3.9 (Profiles approximate interval indicators). *For every $c \in [1, 2]$ and $\epsilon > 0$, there is an averaging profile φ with*

$$\|\varphi - h_c\|_{L^1(\mathbb{R})} < \epsilon, \quad h_c = c^{-1} \mathbf{1}_{(0,c]}.$$

Proof. Let ρ be the mass-one bump constructed above, and set $\rho_\delta(t) = \delta^{-1} \rho(t/\delta)$. For sufficiently small $\delta > 0$, the real smooth function $v = h_c * \rho_\delta$ has mass one and $\|v - h_c\|_1 < \epsilon/2$. Indeed,

$$\|h_c(\cdot - s) - h_c\|_1 = 2 \min(|s|, c)/c,$$

and averaging this identity against ρ_δ proves the claim. Also $\|v^{(n)}\|_\infty \leq \delta^{-n-1} \|\rho^{(n)}\|_\infty$, so v satisfies (3.2). Fix v , and write $m_j = \int t^j v(t) dt$ for $1 \leq j \leq 3$.

Choose four distinct real numbers z_0, z_1, z_2, z_3 , and define

$$P_j(z) = \int_{\mathbb{R}} (z+u)^j \rho(u) du, \quad \rho_{i,A}(t) = A^{-1} \rho(t/A - z_i) \quad (A \geq 1).$$

The j th moment of $\rho_{i,A}$ is $A^j P_j(z_i)$. Since P_j is monic of degree j , the fixed matrix $M = (P_j(z_i))_{j,i=0}^3$ is an invertible triangular change of the Vandermonde matrix $(z_i^j)_{j,i=0}^3$. Solve

$$M\beta = (0, -m_1/A, -m_2/A^2, -m_3/A^3)^\top.$$

The inverse matrix is fixed, hence $\sum_i |\beta_i| = O_v(A^{-1})$. Thus

$$\varphi = v + \sum_{i=0}^3 \beta_i \rho_{i,A}$$

has mass one, its first three moments vanish exactly, and its added L^1 error is $O_v(A^{-1})$. Take A large enough that this error is less than $\epsilon/2$. Finite addition and fixed dilations preserve (3.2), so this real compactly supported function is the required profile. $\square$

Completion of the proof of Theorem 1.1. Continue with $|f_j| \leq 1$, and write

$$A_N(x) = \frac{1}{N} \sum_{n=1}^N F_n(x), \quad p = \prod_{j=1}^3 \int_X f_j d\mu.$$

For every rational $c \in [1, 2]$ and integer $h \geq 1$, choose an averaging profile $\varphi_{c,h}$ with $\|h_c - \varphi_{c,h}\|_1 < 2^{-h}$. The preceding results give a measurable conull set on which $V_k^{\varphi_{c,h}} \rightarrow p$. Intersect these sets over the countable family, obtaining a conull set E .

For $N_k = \lfloor c2^k \rfloor$, the choice of endpoints in h_c gives the exact identity

$$2^{-k} \sum_{n \in \mathbb{Z}} h_c(n/2^k) F_n(x) = \frac{N_k}{c2^k} A_{N_k}(x).$$

Since $|F_n| \leq 1$, for every profile φ ,

$$(3.16) \quad |A_{N_k}(x) - V_k^\varphi(x)| \leq \frac{1}{c2^k} + 2^{-k} \sum_{n \in \mathbb{Z}} |h_c(n/2^k) - \varphi(n/2^k)|.$$

The second term tends to $\|h_c - \varphi\|_1$: its summand is a bounded compactly supported Riemann-integrable function, with possible discontinuities only at $0, c$. Apply (3.16) with $\varphi_{c,h}$, then let $h \rightarrow \infty$. For every $x \in E$,

$$(3.17) \quad A_{\lfloor c2^k \rfloor}(x) \longrightarrow p \quad \text{for every rational } c \in [1, 2].$$

Fix an integer $s \geq 1$, and use the finite rational grid $c_i = 1 + i/s$, $0 \leq i \leq s$. If $2^k \leq N < 2^{k+1}$, choose the largest i such that $n = \lfloor c_i 2^k \rfloor \leq N$. Then $0 \leq N - n \leq 2^k/s + 1$. Averages of terms bounded by one satisfy, whenever $1 \leq n \leq N$,

$$|A_N - A_n| \leq \frac{2(N - n)}{N}.$$

By (3.17), the maximum error over the finite grid tends to zero for $x \in E$. Consequently

$$\limsup_{N \rightarrow \infty} |A_N(x) - p| \leq 2/s \quad (x \in E).$$

Let $s \rightarrow \infty$ to obtain convergence along all positive integers.

Finally undo the normalization of the functions and use $S = T^g$, $b_j = \lambda_j/g$, as in the initial reduction. If the functions are specified only as essentially bounded classes, choose bounded representatives and remove the union of all integer translates of their exceptional sets. This is still a measurable null set. Every other exceptional set used above is measurable and every intersection is countable. Thus the proof needs no standardness or completeness assumption on the probability space. $\square$

4. QUADRATIC STRUCTURE AND LOCALIZATION FOR THE NEW SLOPES

We begin the proof of Theorem 2.2 by establishing the quadratic detection and localization estimates for the configuration $y_j = x + b_j t$. These estimates supply the input to the short-block and frequency arguments of Section 5. The relevant constructions are those of [H, Sections 3–6] and [V, Sections 4–6]. We retain their interval projections and stopping constructions, and prove below the coefficient-dependent identities and estimates on which those constructions rely.

Throughout this section, the slopes are fixed. Constants may depend on them and on the admissibility parameters. A rational number has height at most H if its numerator and positive denominator in reduced form are at most H in absolute value; matrix heights are entrywise. We use the quantitative convention of [H, Section 3.1]: a quasipolynomial bound in parameters $H_1, \dots, H_m \geq 2$ is a bound of the form

$$\exp \left(C_{q,b} \left(1 + \sum_{a=1}^m \log H_a \right)^C \right).$$

The degree C is chosen independently of q , the number of depths, and the input schedules. A fixed number of such constructions may be composed. Constructions repeated a variable number of times retain the explicit count estimates in H and V.

4.1. Local bounds and the chart class. For a scalar or Hilbert-valued function, write $\|z\|_{p,I} = (|I|^{-1} \int_I \|z(y)\|^p dy)^{1/p}$, with the essential-supremum convention for $p = \infty$. Norms without an interval subscript use ordinary Lebesgue measure. The elementary bound used throughout the argument is unchanged except for its constant.

Lemma 4.1 (Local absolute bound). *If $1 \leq p_j \leq \infty$ and $\sum_j p_j^{-1} \leq 2$, then*

$$|I|^{-2} \int_{\mathbb{R}^2} |w_I(x, t)| \prod_{j=0}^3 \|z_j(x + b_j t)\| dx dt \leq C_b \prod_{j=0}^3 \|z_j\|_{p_j, I}.$$

The assertion also holds for Hilbert-valued inputs, using their norms in the integrand.

Proof. For two distinct roles i, j , the map $(x, t) \mapsto (y_i, y_j)$ has determinant $b_j - b_i \neq 0$. This proves the endpoint with $p_i = p_j = 1$ and the other exponents infinite. The support and kernel bounds give the all-infinite endpoint, and normalized norm monotonicity gives the endpoints

with only one exponent equal to one. Log-convexity of the positive integral fills the stated polytope, as in [H, Lemma 2.1]. $\square$

On the support of a local kernel, all four coordinates lie in an interval of length $r = |I|$. Consequently

$$(4.1) \quad |t| \leq \frac{r}{\max_j b_j - \min_j b_j} \leq \frac{r}{3}.$$

In any pair of coordinate variables, the rescaled kernel thus has support in a fixed box and the same type of derivative bounds. Its integral in either coordinate with the other fixed is zero, by the corresponding one-coordinate marginal hypothesis. These facts also give $C_N(q) \lesssim_b N$ and preserve the normalization-removal argument of [H, Lemma 2.4].

We recall the concrete class in which the quadratic structure is detected. Write $e(u) = \exp(2\pi i u)$. A *quadratic chart atom* is a function

$$(4.2) \quad \phi(y) = e(\lambda y^2 + \beta y) \sum_{n \in \mathbb{Z}^d} \chi(\theta y - n) e((\theta y)^T C(\theta y - n)),$$

where $\theta \in \mathbb{R}^d$, $\lambda, \beta \in \mathbb{R}$, $C \in \mathbb{Q}^{d \times d}$, and χ is smooth and compactly supported. An overall constant of modulus one is allowed. We use exactly the complexity convention of [H, Definition 4.1]: for complexity $H \geq e^2$, $d \leq \log H$, the height of C is at most H , $\text{supp } \chi \subset [-H, H]^d$, and

$$\|\partial^\alpha \chi\|_\infty \leq H^{c_{\text{ch}}(|\alpha|+1)} (|\alpha| + 1)^{c_{\text{ch}}(|\alpha|+1)}$$

with a fixed chart constant c_{ch} . The real oscillation parameters λ, β, θ are unrestricted. A bounded atom has supremum norm at most one; any normalization factor is put in its external scalar coefficient.

The elementary operations, Weyl alternative, and chart expansion of [H, Lemmas 4.2–4.4] concern this class itself. Their proofs therefore apply without a coefficient change. We will need the precise splitting supplied by the Weyl alternative. In its multivariable version replace θy by Vz , and the ordinary quadratic phase by $z^T \Lambda z$, where Λ is real symmetric and z has at most three coordinates. Put

$$C_{\text{sy}} = \frac{1}{2}(C + C^T), \quad C_{\text{sk}} = \frac{1}{2}(C - C^T), \quad \text{sym } M = \frac{1}{2}(M + M^T).$$

Let $Q(z)$ be this multivariable chart. Suppose $0 < \delta < 1/2$ and a Lipschitz function ϑ , supported in a fixed box with supremum and Lipschitz norms at most H , satisfies

$$\sup_\xi \left| \int \vartheta(z) Q(z) e(-\xi^T z) dz \right| \geq \delta.$$

The Weyl alternative then supplies a rational subspace U , the orthogonal projection P_f onto U , $P_s = 1 - P_f$, and the bounds

$$(4.3) \quad \begin{aligned} (C - C^T)|_{U \times U} &= 0, & V_f &= P_f V, & V_s &= P_s V, & \|V_s\| &\leq H', \\ \Lambda_* &= \Lambda + \frac{1}{2} V^T C_{\text{sy}} V + \text{sym}(V_f^T C_{\text{sk}} V_s), & \|\Lambda_*\| &\leq H'. \end{aligned}$$

The projection heights are at most H' , and H' is quasipolynomial in H, δ^{-1} . The subsequent chart expansion has the absolute coefficient and pooled Fourier-tail bounds of [H, Lemma 4.4]. These are the bounds used below, including when expansions must be summed over depths.

4.2. Separation along the configuration and local detection. The first coefficient-dependent step moves two structured inputs into the other two coordinates and the step variable. This is the form of separation needed for both detection and the later predictor estimates.

Lemma 4.2 (Progression separation). *Let i, k, a, l be the four distinct roles and let ϕ_i, ϕ_k be bounded chart atoms of complexity at most H . On $y_j = x + b_j t$, their product has an absolutely convergent expansion*

$$\phi_i(y_i)\phi_k(y_k) = \sum_{\nu} a_{\nu} \psi_{a,\nu}(y_a) \psi_{l,\nu}(y_l) Q_{\nu}(t)$$

with bounded chart atoms as factors and the same shell bounds as [H, Lemma 4.5], with constants depending on b . In particular, for a quasipolynomial $D = D_b(H)$, a dyadic Fourier shell of size $T \geq 1$ has absolute coefficient sum at most DT^{-100} , and its factors have complexity at most $(DT)^C$.

Proof. First separate ϕ_i , temporarily suppressing its linear phase. Choose the fixed rationals

$$\alpha_i = 1, \quad \alpha_k = 0, \quad \alpha_a = \frac{b_i - b_l}{b_l - b_a}, \quad \alpha_l = \frac{b_a - b_i}{b_l - b_a}.$$

They satisfy $\sum_j \alpha_j = \sum_j b_j \alpha_j = 0$. Insert the smooth integer partitions of unity from H at $\theta y_a, \theta y_l, \theta t$, and write the lifts as $d_j = \theta y_j - m_j$ and $d' = \theta t - m$. For $j = i, l$, set

$$k_j = d_j - d_a - (b_j - b_a)d', \quad k_a = 0.$$

These carries are integer vectors. The support restrictions bound them by the chart budget, with a fixed slope factor. Conversely, k_i, m_a, m determine m_i uniquely. Each remaining specified carry equality is imposed exactly by a smooth cutoff equal to one at the specified integer vector and zero at all other integer vectors.

For $S_{\alpha} = \sum_j \alpha_j b_j^2$, direct expansion gives

$$(4.4) \quad \begin{aligned} \sum_j \alpha_j (\lambda y_j^2 + (\theta y_j)^T C d_j) &= S_{\alpha} (\lambda t^2 + (\theta t)^T C d') \\ &\quad + \sum_j \alpha_j (\theta y_j)^T C k_j. \end{aligned}$$

Indeed, substituting $d_j = d_a + (b_j - b_a)d' + k_j$ cancels the common-lift and x -terms. Here $S_{\alpha} = (b_i - b_a)(b_i - b_l)$. Solving for the phase in role i leaves chart matrices $-\alpha_a C, -\alpha_l C, S_{\alpha} C$ in the three desired variables. For fixed carries, the last sum is linear in x, t , and hence splits between y_a, y_l ; so does the suppressed linear phase.

The coupled amplitude is smooth in d_a, d_l, d' . Its Fourier expansion on a larger fixed box separates those variables, exactly as in the proof of [H, Lemma 4.5]. The new substitutions cost at most C_b^n in derivatives of order n , and their rational heights grow by fixed factors. The possible integer choices increase by at most $C_b^{O(d)}$. Since $d \leq \log H$, these changes are within the quasipolynomial chart budget. H's integration by parts, with derivative order exceeding the Fourier dimension by at least 110, gives the stated absolute shell bound. Separate the second atom with $\alpha_i = 0$ and combine the factors by direct sums of horizontal coordinates. This completes the expansion. $\square$

For one separated term, let

$$A_r = \sup_{\xi \in \mathbb{R}} \left| \int_{\mathbb{R}} Q(ru) \zeta(u) e(-\xi u) du \right|,$$

where ζ is smooth, nonnegative, equal to one on $[-1/3, 1/3]$, and has $\|\zeta\|_1 \leq 1$, as in [V, Section 5.4]. The two-input form in the remaining roles has norm at most $C_b A_r$ on normalized $L^2(I)$ inputs. To see this, use

$$t = \frac{y_l - y_a}{b_l - b_a}.$$

Fourier-expand the smooth normalized kernel in those two coordinates. Each term is a modulated convolution with a fixed dilation of $Q(t)\zeta(t/r)$, so Plancherel gives the claimed bound. If $A_r \geq \delta$, the Weyl alternative and chart expansion further give the finite smooth separation in [H, Lemma 4.6]: for every $0 < \varepsilon < 1/2$, the two-coordinate kernel has a separated approximation with uniform error at most ε , whose number of terms, atom complexities, and absolute coefficient sum are quasipolynomial in $H, \delta^{-1}, \varepsilon^{-1}$. The retained phases are $\lambda_* t^2 + \gamma t$, with $|\lambda_*| r^2$ bounded by the stated budget, and the linear phase splits as

$$e(\gamma t) = e\left(-\frac{\gamma y_a}{b_l - b_a}\right) e\left(\frac{\gamma y_l}{b_l - b_a}\right).$$

Thus the fixed rational-height description of the effective quadratic and linear frequencies in that lemma is preserved.

Lemma 4.3 (Local detection). *Fix $q > 2$ and $C_1 \geq 1$. Suppose that the four inputs R, h, u, v , in any slot order, satisfy*

$$\|R\|_{2,I} \leq C_1, \quad \|h\|_{q,I} \leq 1, \quad \|u\|_{2,I}, \|v\|_{2,I} \leq 1, \quad |H_I(R, h, u, v)| > \delta,$$

where $0 < \delta < 1/2$. There is a bounded chart atom ϕ of complexity at most D such that

$$\left| \frac{1}{|I|} \int_I R(y) \overline{\phi(y)} dy \right| \geq D^{-1}, \quad D \leq \exp\left(C_{q,b}(1 + \log C_1 + \log(1/\delta))^C\right).$$

The degree C is independent of q .

Proof. We verify the progression estimate in [H, Lemma 4.11] and then apply its detection proof. Normalize $I = [0, 1)$. Extend the kernel to the same fixed circle used there. By (4.1), its real support fits in the periodic (x, t) -box, and all real y_j on that support already belong to I . Hence zero-extended inputs on the circle give the correct original integral, even when some slopes are large or negative. The kernel's absolute Fourier coefficient sum is uniformly bounded.

The integer left inverse in Lemma 2.3 distributes a circle character with indices (p, q') into slot characters with indices $p\mathbf{1}_{j=0} + q'e_j$. To a character-weighted progression average apply H's three Cauchy–Schwarz steps, taking Haar differences in the directions $(-b_j, 1)$. The last cube has three independent increments multiplied by nonzero differences $b_i - b_j$. Each such integer multiplication pushes circle Haar measure to itself. The last integral is therefore the eighth power of the same circle U^3 norm as in H. Character multiplication preserves that norm, the L^2 norm, and the supremum norm. This proves H's local generalized von Neumann estimate; its accompanying two- L^1 bound follows from Lemma 4.1.

The continuous inverse theorem and bounded-input decomposition in [H, Sections 4.5–4.6] concern functions on this circle and its chart atoms, and require no change. In the proof of [H, Lemma 4.12], clip the three opposing inputs at height B . The local absolute bound gives total error at most $C_b C_1 B^{-(q-2)/q}$. Taking B of order $(C_b C_1 / \delta)^{q/(q-2)}$, apply that bounded-input decomposition twice. The error estimates use the generalized von Neumann and two- L^1 bounds just proved. Two bounded atoms remain opposite R and an input of normalized L^2 norm at most one. Lemma 4.2 and the preceding single-scale separation give a finite sum of products of one-coordinate moments, with controlled absolute coefficient sum. One moment of R consequently has the asserted size. These are a fixed number of quasipolynomial constructions.

The clipping exponent contributes only a q -dependent multiplier to a logarithm, so the final logarithmic degree is independent of q , as required. $\square$

4.3. The base reduction and inputs that vary with depth. Detection is used in H and V to construct finite approximation spaces on interval trees. We recall their meaning before verifying the scale estimates. A column with label v , born on an interval J_v , is an augmented function

$$\Phi_v(y) = (\phi_v(y), \sqrt{\epsilon_v} \mathbf{u}_v), \quad y \in J_v,$$

where ϕ_v is a scalar chart atom, $\epsilon_v > 0$, and the $\mathbf{u}_v$ are distinct orthonormal private coordinates. Only the first, physical coordinate enters H_I . A local projection on I is the orthogonal projection onto constant-coefficient combinations of the available columns restricted to I . Lists are inherited on descendants. Accordingly the direct-sum projections on successively finer interval partitions have nested ranges. The private-coordinate, compression, and height-bin estimates of [H, Section 3] depend only on these definitions and the column budgets, and apply unchanged.

There are two coefficient-dependent scale estimates in the base construction. For two fixed chart atoms, the separation above reduces the other two inputs to a kernel of the form $w_I(x, t)Q(t)$, with fixed bounded one-coordinate multipliers. The proof of [H, Lemma 5.1 and Corollary 5.2] expands Q , groups nearby linear frequencies, and uses the zero marginals of w_I . In coordinates (y_a, y_l) , its double primitive has compact support in an interior rectangle. Fourier expansion followed by one derivative in each coordinate expresses it as a sum of tensor products of mean-zero smooth bumps. In role $p \in \{a, l\}$, the resulting scalar rows have fixed bounded carrier functions

$$m_{\nu,p}(y) = e(\nu y) \psi_p(y), \quad \nu = \pm \frac{\gamma}{b_l - b_a},$$

where ψ_p is the fixed separated multiplier in that role. These are the only coefficient replacements in the row estimate. They alter frequency separations and rational heights by fixed factors. H 's height-bin argument, ordinary Bessel estimate, and compression step therefore give its absolute scale sum and finite predictor list with the same quantitative orders, including the pooled-tail bound needed for summation over all depths. The finer partition comparison in [H, Lemma 5.3] uses the resulting one-coordinate predictor moments, so its projection-increment proof applies with the same replacement.

For the second estimate, fix three bounded chart columns and take the transpose in role l at $y + h$. Its other coordinates are

$$y + h + (b_j - b_l)t \quad (j \neq l).$$

After squaring the normalized L^2 norm, the six atoms form a multivariable chart in (h, t, t') . At a fixed path point y , all parameters before multiplication by r or r^2 remain fixed as the depth changes. H 's Weyl alternative, chart expansion, and height-bin argument therefore apply. Away from the bounded number of intermediate scales, the bounded effective quadratic parameters are tiny and each retained t -frequency is tiny or large. Smoothness handles the large frequencies. Removing the tiny oscillations leaves exactly

$$\int_{\mathbb{R}} w_I(y + h - b_l t, t) dt = 0.$$

This proves the active-scale count and the small-norm scale sum in [H, Lemma 5.7], with constants depending on b .

For scheduled inputs we use [V, Section 4]. Its row estimates concern Hilbert-space operators on refining interval partitions, evaluated on inputs at prescribed nondecreasing read depths. Their hypotheses are support on the input partition, representation of the adjoint ranges on the observation partition by the inherited columns, the stated order of successive partitions,

and an ordinary Bessel bound before evaluation on the varying inputs. None involves the maps y_j . Restriction to a new spatial root preserves the pointwise block bound in Theorem 2.2. The original W_j and original depth schedules are retained; subsequences may have empty intersections with some blocks, as allowed in V's row argument. Its nearest-endpoint extension is retained as well.

The smooth rows in [V, Lemma 4.6] require the ordinary Bessel bound just verified, with the fixed carriers above. Thus the scheduled scale-sum and predictor results [V, Lemma 5.3 and Corollary 5.4], and its delayed replacement Lemma 5.5, hold with the new coefficients. The remaining steps of its Section 5 use the same local detection, fixed-cell fits, projection energies, and staggered pair comparisons. A finer projection in a comparison is still applied to the input at the prescribed read depth, and its row estimate has no factor depending on W_j .

We now describe the output of the base construction. For the fixed-input argument, embed the scalar input as $F_{j,s} = (f_j, 0)$, independent of s , on a subtree satisfying $\|f_j\|_{q,I} \leq 1$ on every cell, as in [H, Proposition 2.3]. For the scheduled argument, put $F_{j,s} = (z_{j,s}, 0)$ with the inputs of Theorem 2.2. The adaptive fits and two-earliest-increments expansion of [H, Lemmas 5.4–5.5], or [V, Sections 5.1–5.2], express the original form, at any finite accuracy cutoff, in tuples with two differences of fitted projections and two original-input residuals, together with a remainder whose absolute length-weighted sum tends to zero on the fixed finite tree. Each tuple has an earlier and a later accuracy index, and a choice of the four roles. Their exact staggered ordering is retained from those results. High-position truncation, using the predictor estimates above, is [H, Proposition 5.6] or [V, Sections 5.5–5.6]. It assigns each original summation cell to a segment root R with inherited projection families; write $\mathcal{I}_R$ for the cells assigned to R . Auxiliary projection cells are not additional summands.

Fix one tuple and let k be its later accuracy logarithm. The two positions called low in this construction are denoted i, j , and the other two by a, l . On a cell of depth s , the resulting arguments satisfy

$$(4.5) \quad \begin{aligned} Y_{p,I} &= B_{p,I} + U_{p,I}, & U_{i,I} &= U_{j,I} = 0, \\ B_{p,I} &= \sum_{h=1}^{m_p} \varepsilon_{p,h} Q_s^{p,h} F_{p,s}, & m_p &\leq 2, \quad \varepsilon_{p,h} \in \{-1, 1\}, \\ U_{p,I} &= (P_{p,s+K_0}^{\text{new}} - P_{p,s}^{\text{new}}) F_{p,s} \quad (p = a, l). \end{aligned}$$

All expressions are restricted to I ; a projection indexed by a finer depth is the direct sum on that finer partition. The $Q^{p,h}$ and P_p^{new} are the inherited basic families just constructed, and K_0 is their positive integer delay. Their column sizes, inverse private weights, scalar complexities, coefficient sums, and path counts have a joint budget $e^k \leq D_{\text{base}} \leq \exp(C_{q,b} k^C)$. On every segment they give

$$(4.6) \quad \begin{aligned} \|B_{p,I}\|_{2,I}, \|U_{p,I}\|_{2,I}, \|Y_{p,I}\|_{2,I} &\leq C_{q,b}, & \Lambda_k &\leq C_{q,b} k^C, \\ \sum_{I \in \mathcal{I}_R} |I| \|U_{p,I}\|_{2,I}^2 &\leq \Lambda_k^2 |R| \quad (p = a, l). \end{aligned}$$

One may take Λ_k to dominate $C_{q,b}(1 + K_0) \log(2D_{\text{base}})$. For the fixed-input argument the smaller bound in H also suffices.

For noninitial accuracy levels, every pair of full Y arguments has form norm at most $\eta = \exp(-k^\theta)$, and $|H_I(Y_I)| \leq \eta$. Here the pair norm is the supremum against normalized $L^2(I)$ inputs in the complementary two positions; $0 < \theta < 1$ is fixed before q . Define $\mathcal{A}_R$ to consist of those cells for which some available triple of basic scalar columns has transpose norm greater than $D_{\text{base}}^{-C_{\text{act}}}$, with the fixed sufficiently large exponent in H or V. The active-scale estimate gives $\sum_{I \in \mathcal{A}_R} \mathbf{1}_I \leq D_{\text{base}}^C$.

There are three remaining patterns: $z^\varnothing = B$, z^a replaces only role a of B by U_a , and z^l replaces only role l by U_l . Put

$$(4.7) \quad \mathfrak{M}_R(z) = \sum_{I \in \mathcal{A}_R} |I| \min\{\eta, |H_I(z_I)|\}.$$

The output of [H, Proposition 5.8] and [V, Proposition 5.6] is

$$(4.8) \quad \sum_{I \in \mathcal{I}_R} |I| |H_I(Y_I)| \leq C_{q,b} \eta \Lambda_k^2 |R| + e^{-k} |R| + \sum_{z \in \{z^\varnothing, z^a, z^l\}} \mathfrak{M}_R(z).$$

Indeed the term with both U factors uses the full complementary pair bound and (4.6). The inactive triple scale sum bounds the omitted cells. On the others, apply $\min(\eta, \sum x_m) \leq \sum \min(\eta, x_m)$ to the full tuple before separating its patterns. No individual pattern is asserted to have the full-pair smallness.

For each fixed tuple the segment roots have total length bounded by a fixed multiple of the original root length, and the sum of the high-truncation replacement errors is at most $C_{q,b} e^{-k}$ times that length. These are the segment-packing assertions in the cited base constructions. The finitely many initial accuracy levels use the active path bound and local sizes, without the factor η , and hence have depth-independent bounds. All these estimates are valid anew on a normalized subroot with its inherited lists and original input schedule. We have therefore reduced both input classes to estimating (4.7) uniformly over these concrete base patterns.

4.4. Curvature tags and separation of distinct tags. We now localize the base arguments according to their quadratic structure. Recall the nonzero integers c_j of Lemma 2.3. A column in role i has *normalized curvature tag* $v = (\theta_v, C_v, \lambda_v)$ when its quadratic chart phase is written as

$$c_i(\lambda_v y^2 + (\theta_v y)^T C_v (\theta_v y - n)).$$

Its linear modulation is not part of its tag. This is H's normalization with $(-1, 3, -3, 1)$ replaced by c . Division by the fixed nonzero c_i preserves the chart budgets.

At length r , the difference data of two tags v, v' are

$$(4.9) \quad V' = r \begin{pmatrix} \theta_v \\ \theta_{v'} \end{pmatrix}, \quad C' = \text{diag}(C_v, -C_{v'}), \quad \Lambda' = r^2(\lambda_v - \lambda_{v'}).$$

The ambient graph in [H, Section 6.1] joins two tags when these data admit (4.3) with bounds at most the chosen edge height H_{ed} ; equal tags have distance zero. This graph includes all hypothetical tags within the basic budget. Its distances depend on tags and depth, and do not increase when the interval length is halved.

Lemma 4.4 (Separation of distinct tags). *For single-column complexities at most $\exp(C_{q,b} k^C)$ and $\log(1/\tau) \leq C_{q,b} k^C$, the edge height can be chosen with $\log H_{\text{ed}} \leq C_{q,b} k^C$ so that tags at graph distance greater than one have both of the following properties. Their same-slot normalized physical inner product is at most τ in absolute value. In distinct slots, their form norm against two arbitrary normalized $L^2(I)$ spectators is at most τ .*

Proof. For the first assertion use the Weyl alternative on the conjugate product, after the smoothing of the interval indicator in [H, Lemma 6.1]. Its fixed nonzero factor c_i changes the allowed rational heights by a fixed amount, and multiplying back gives the splitting for the tag difference.

For distinct source slots i, j , let the spectators be at a, l , and write the centered source coordinates in terms of the normalized spectator coordinates z, w . Their linear parts are

$r(\xi_p z + \zeta_p w)$, where

$$\xi_p = \frac{b_l - b_p}{b_l - b_a}, \quad \zeta_p = \frac{b_p - b_a}{b_l - b_a} \quad (p = i, j).$$

These are nonzero fixed rationals. The cancellation of polynomials of degree at most two gives

$$(4.10) \quad c_i \xi_i \zeta_i = -c_j \xi_j \zeta_j =: \kappa \neq 0.$$

Use the difference data (4.9). For a small return $V'h = m + d_h$, choose an integer period Π and require $m \in \Pi \mathbb{Z}^{d_v + d_{v'}}$, where $d_v, d_{v'}$ are the horizontal dimensions of the two tags. Choose Π sufficiently divisible that $\xi_i \Pi, \xi_j \Pi$ are integers and $\Pi C_v, \Pi C_{v'}, c_i \xi_i \Pi C_v, c_j \xi_j \Pi C_{v'}$ have integral entries. This clears the lattice-index and phase denominators together. Translate z by h , and the source lift indices by $\xi_i m_v$ and $\xi_j m_{v'}$, respectively. The phase-increment calculation in the proof of [H, Lemma 4.3], together with (4.10), gives the modulation in w

$$\kappa \Omega(h, m), \quad \Omega(h, m) = (2\Lambda' + (V')^T C' V')h - (V')^T (C' - (C')^T)m.$$

Indeed each source contribution has the multiplier $c_p \xi_p \zeta_p$; the two contributions therefore combine into these difference data. The remaining linear terms in z are absorbed in the first spectator. The amplitude and kernel errors are bounded by a quasipolynomial times $|h| + \|d_h\|$, as in H.

Call the two spectators f, g , and clip them at a sufficiently large inverse polynomial in τ . Their discarded L^1 tails are small by their L^2 bounds and Lemma 4.1. If the form is larger than τ , applying the return calculation to sufficiently many multiples of (h, m) gives large correlations of one fixed bounded kernel with tests $f_\ell(z)g_\ell(w)$, where the f_ℓ have bounded L^2 norms and

$$g_\ell(w) = g(w)e(\ell \kappa \Omega(h, m)w)$$

up to a common bounded multiplier. The small return threshold is chosen before these multiples so that all translation errors remain small. No regularity of the first spectator is required: it is translated by a change of variables. If $|\Omega(h, m)|$ exceeds a fixed constant, the second tests are Bessel on the fixed box, with bound controlled by the clipping height. Cauchy–Schwarz in z then contradicts the number of large correlations. Thus Ω is bounded on every sufficiently small return.

The concluding return-to-splitting argument in the proof of [H, Lemma 4.3] uses precisely this bound: its alternating return identity yields an isotropic lattice space, successive minima bound the rational fast projection and slow part, and coordinate-direction returns bound the effective quadratic form. The enlarged denominator-clearing period changes only its quasipolynomial constants. It supplies (4.3), contradicting the assumed graph distance. This is the distinct-slot argument of [H, Lemma 6.1] for the present coefficients. $\square$

4.5. Same-tag tests and the localization output. The complementary fact is that a column of one tag can detect a residual in another role using a template of that same tag. This keeps the approximation spaces finite without repeatedly increasing their nonlinear complexity. A pure source column means one scalar column label, rather than a linear combination of labels.

Lemma 4.5 (Uniform same-tag test). *Fix an original tag v and its original source columns within the complexity budget $\exp(C_{q,b}k^C)$. There are fixed template families in all four slots, consisting of bounded columns of tag v with arbitrary external linear modulations. Suppose a pure source column from the original or template family in role i has form norm greater than $t_0 > 0$ against a physical residual in $L^2(I)$ in role j and two normalized $L^2(I)$ spectators. The residual then has a normalized moment of magnitude at least t_0/H_0 against a template in role j , where $H_0 \leq \exp(C_{q,b}k^C)$. This bound is independent of the external linear modulations and of the number of earlier template insertions.*

Proof. Use the nonnegative partition function ρ and template amplitudes $\rho(d)e(\ell^T d/P_0)$ of [H, Lemma 6.6], with ℓ ranging over residue vectors modulo a sufficiently large integer period P_0 . For the four lifts $d_p = \theta_v y_p - n_p$, set

$$(4.11) \quad d_\diamond = \sum_p e_p d_p, \quad k_p = d_p - d_0 - b_p d_\diamond.$$

The Bézout identities show that $d_\diamond$ is a lift of $\theta_v t$ modulo the integer lattice and every k_p is integral. All are bounded on the branch boxes. The common quadratic phases cancel by Lemma 2.3; for fixed carries the remaining phase is

$$\sum_p c_p (\theta_v y_p)^T C_v k_p,$$

which is linear in the coordinates. Smooth equality cutoffs in the four lifts impose (4.11) exactly, because their arguments are integer valued on the configuration. Their Fourier expansion on a sufficiently large box therefore produces the same template amplitudes. Increasing the period and the possible carry count costs only the allowed chart budget.

As in H, write each Fourier index as $\ell = P_0 q' + r'$. The factor $e((q')^T d_p)$ is an external linear modulation, so only r' enters the template amplitude. This gives the same bound even when the source was itself a previously inserted template. For an original source, the remaining smooth horizontal amplitude, after phase cancellation, is periodized and expanded in ordinary torus characters. Its remaining dependence on the source coordinate is a linear modulation, which is absorbed into the spectator tests. All these absolute coefficient sums are bounded by the same quasipolynomial budget. After separating the smooth kernel, express the target coordinate as $y_j = \alpha y_a + \beta y_l$, with fixed nonzero rational α, β . Fourier-expand the demodulated residual on a fixed larger interval. If M is the supremum of its normalized template moments, one separated term is bounded by

$$M \sum_m |A_m B_m| \leq M \left(\sum_m |A_m|^2 \right)^{1/2} \left(\sum_m |B_m|^2 \right)^{1/2} \leq C_b M.$$

The last inequality is Bessel's inequality for spectator frequencies with nonzero spacings proportional to α, β . Summing the absolute Fourier coefficients gives $H_0 M$, and proves the assertion. $\square$

We record how these two tests enter the remainder of the imported construction. This also fixes the meaning of its output used in Section 5. Put $P = \lceil \eta^{-\alpha} \rceil$, where $\alpha > 0$ will be chosen in the closing argument. The vertices are occurrences of the original columns and are ordered by admission. Added main and helper columns attach to those vertices and have their normalized tags; they create no new graph vertices. H's short-path, color, and historical-mask constructions [H, Lemmas 6.2–6.5] form groups of these vertices and assign current groups to each summation cell. An actual use is such an occurrence of a current group at a cell. A mask is the subset of available vertices allowed in a specified local projection. A pad enlarges that subset by the prescribed graph radius; a positive smeared filter is the average of P nested pad projections. A history is the ordered product of such filters or their complements used in the binary localization. The number $\mathcal{L}$ of colors is polynomial in k , and these main histories have $O(\log \mathcal{L})$ factors. Each group records a distinguished tag z , the primary winner in H's color construction. Every vertex admitted to its masks has, already at admission, a graph path to z of length polynomial in P, k . The short-path lemma therefore gives the endpoint splitting (4.3) relative to z , with logarithmic height polynomial in P, k . This is the splitting used for the first matching of a column in Section 5.

We use the precise masks and historical-use convention of those lemmas. Its relevant properties are as follows. Masks are inherited on descendants; at an actual use, enlarging a pad by one includes every neighbor of the preceding mask. An endpoint is usable when its mask is nonempty and its cell has a downstream actual use of its group. Distinct groups of one color have tags separated by graph distance greater than one on overlapping usable endpoints. The endpoint bound along short paths follows from the rational splitting (4.3). The usable-space Bessel bound follows from same-slot separation in Lemma 4.4. These proofs involve only rational linear algebra, inherited interval projections, and the indicated splitting and separation bounds; thus their path, rank, and pad counts keep the quantitative orders in H.

For completeness, the closure condition imposed on these spaces has an exact form. At rank r , let n_{r-1} bound the number of labels through the preceding rank along a path, let $D = D_{\text{base}}$, and let $\tau_3 = D^{-C_\tau}$. For a basic argument a , each prescribed history $\mathcal{H}$, target mask $\mathcal{M}$, and pure lower-rank source whose tag belongs to that mask, the form norm against the residual $(1 - P_{r,\mathcal{M}})\mathcal{H}a$ and two normalized L^2 spectators is required to be at most

$$(4.12) \quad \frac{\tau_3}{D\sqrt{n_{r-1}}} \|a\|_{2,I}.$$

On failure, Lemma 4.5 supplies a new column and hence a projection increment. The products-of-prefixes estimate of H Section 3 bounds the sum of those increments, exactly as in its Lemma 6.7. Its rank recurrence and main/helper catalog counts are therefore unchanged. This count depends on the names of masks and histories, including anticipated helper bases, but not on the numerical edge height or on later mode precision. As in [H, Section 6.4] and [V, Section 6.1], first fix these counts, then the mismatch tolerance, and finally the edge height needed in Lemma 4.4. The lower-rank count in (4.12) is retained, so no full source count is lost when these errors are summed.

It follows that the binary localization and outgoing-filter identities of [H, Proposition 6.8 and Lemma 6.9] hold for the present forms: they use (4.12), distinct-tag form separation, the local absolute bound, and the same nested projection algebra. In the color construction of [H, Lemma 6.3], the first failure of its marking conditions may remove a subtree. Increasing the fixed thresholds makes the sum of these roots' lengths an arbitrarily small fixed fraction of $|R|$. Section 6 restarts the original summands on those roots. After these marking-failure cuts, let $\mathcal{A}_R^{\text{main}}$ denote the remaining active cells. For each base pattern and $I \in \mathcal{A}_R^{\text{main}}$ of depth s , the localization is

$$(4.13) \quad \begin{aligned} H_I(z_I) &= \sum_{\lambda} \sum_{G \text{ current}} H_I(Z_{0,G,s}^\lambda, \dots, Z_{3,G,s}^\lambda) + e_I, \\ Z_{p,G,s}^\lambda &= \bar{L}_{p,G,s}^\lambda \mathcal{B}_{p,s}^\lambda z_{p,I}, \quad \sum_{I \in \mathcal{A}_R^{\text{main}}} |I| |e_I| \leq C_{q,b} e^{-k} |R|. \end{aligned}$$

Here λ ranges over polynomially many fixed binary search leaves. The inner history $\mathcal{B}_{p,s}^\lambda$ is independent of G , and $\bar{L}_{p,G,s}^\lambda$ is the outer ball average. The minimum in (4.7) is distributed across the fixed leaves, while the group sum remains inside the original cell contribution. The filters act on the whole basic arguments in this identity. We next record their endpoint bounds, then the algebra that moves the complementary filters onto a side operand.

For varying inputs, this localization acts at each cell on its one actual tuple at that cell's original depth. The insertion tests freeze that operand while the column list grows. The products-of-prefixes estimate therefore remains a fixed-vector estimate, and gives the same closure count; this is the argument of [V, Section 6.1]. After the operator families have been fixed, its Lemma 6.1 and Corollary 6.2 control their changes between endpoint depths. More explicitly, for one slot, search leaf, and color, write the full localized endpoint output as $Z_{G,s} = \bar{L}_{G,s} \mathcal{B}_s B_s$, where $\bar{L}$ is

the outer positive ball average and $\mathcal{B}$ its group-independent inner history. If depth intervals $[r, t]$ have edge-overlap multiplicity at most a , and $E_{G,r,t}$ is a union of t -cells with downstream actual uses certifying the indicated group, those results give

$$(4.14) \quad \sum_{G,[r,t]} \|\mathbf{1}_{E_{G,r,t}}(Z_{G,t} - Z_{G,r})\|_2^2 \leq aG_0|R|, \quad G_0 = \exp(C_{q,b}(1 + \log(2k))^C).$$

The shared histories without an outer ball satisfy the corresponding estimate counted once across groups. Endpoints before the attempt root are zero, and all other endpoints use inherited continuations and local refits. The proof first establishes the ordinary Bessel bounds for operator changes, using usable endpoint ranges and compression, and then evaluates the rows at their prescribed depth inputs. Input changes use only the original block variation bound. Thus the constants in (4.14) are independent of the partitions and W_j .

4.6. Transferring the complementary filters. The side estimate will encounter two distinguished operands while the other two retain their whole localized histories. We state the projection identity for this situation explicitly. Fix an actual group use at a cell I , one search leaf, and four distinct roles a, a', c, d . Let $S_a, S_{a'}$ be fixed Hilbert-valued $L^2(I)$ operands in the first two roles. They are called side operands here because they are retained while the complementary histories are simplified; their particular construction and cross-depth bounds are given in Section 5. The other two roles have the whole localized arguments with basic inputs B_c, B_d from (4.13). Choose the side operands before closing the helper catalog. The helper version of (4.12) is imposed with these fixed states as bases and lower-rank main columns as sources, as in [H, Lemma 6.7].

Expand each complementary factor $1 - \bar{Q}$ as the difference of the identity and a positive average. There are only $2^{O(\log \mathcal{L})}$ resulting products. In one such product, remove the largest-rank outermost positive filter in roles c, d . The positive main ranks are distinct in the source construction, so the remaining source in the other complementary role has strictly lower rank. Suppose the filter being removed is on role c . Let $Q_{c,p}$, $1 \leq p \leq P$, be its individual pad projections, and $Q_{a,p}$ the helper-rank projections in role a with the same masks and pads. For $j = a, c$, set

$$(4.15) \quad \begin{aligned} Q_{j,0} &= 0, & Q_{j,P+1} &= 1, & A_j^p &= Q_{j,p} - Q_{j,p-1} \quad (1 \leq p \leq P+1), \\ \bar{Q}_j &= \frac{1}{P} \sum_{u=1}^P Q_{j,u} = \sum_{p=1}^{P+1} \frac{P-p+1}{P} A_j^p. \end{aligned}$$

The A_j^p are mutually orthogonal projections and sum to the identity. Their final band is included, with weight zero in the positive average.

Let x_c be the inner history on B_c after deleting the outgoing filter, and let x_d be the entire remaining history on B_d . Write $x_a = \mathcal{D}_a S_a$, where $\mathcal{D}_a$ is the product of helper averages already transferred, and $x_{a'} = S_{a'}$. In the next displays arguments are written in role order a, a', c, d ; each still occupies its original slot in H_I . Multilinearity and (4.15) give

$$(4.16) \quad \begin{aligned} &H_I(x_a, x_{a'}, \bar{Q}_c x_c, x_d) - H_I(\bar{Q}_a x_a, x_{a'}, x_c, x_d) \\ &= \sum_{p,q=1}^{P+1} \frac{p-q}{P} H_I(A_a^p x_a, x_{a'}, A_c^q x_c, x_d). \end{aligned}$$

Indeed the coefficient of the pair of bands is the weight of band q in role c minus the weight of band p in role a . This proves H's Equation (15), including its terminal bands, for the present form without a coefficient change.

The estimate accompanying [H, Lemma 6.9] permits only $|p - q| \leq 2$ to be retained: the other terms at this cell and group have total absolute value at most

$$D_{\text{base}}^{-C_3} \|S_a\|_{2,I} \|S_{a'}\|_{2,I}$$

for any prescribed fixed C_3 , after the numerical accuracies are chosen. Its hypotheses are precisely the closure and separation estimates proved above. If $p > q + 2$, expand the lower c -band into pure main sources and use the helper closure test on the higher a -band, written as a difference of residual projections. If $q > p + 2$, expand the lower-rank source in role d . A source tag within one edge of the lower helper mask belongs, by neighbor closure, to the next outgoing mask; the main closure test then controls the higher c -band. For a tag farther away, expand the helper primal and use Lemma 4.4. The coefficient counts are canceled by the lower-rank denominator in (4.12), except for fixed count powers already included when the mismatch tolerance was chosen. The $O(P^2)$ band pairs and polynomially many transfer positions fit those accuracy choices because $\log P = O(k)$.

For each fixed offset $h = q - p \in \{-2, -1, 0, 1, 2\}$, retain the coupled sum

$$(4.17) \quad \mathcal{B}_{I,h} = -\frac{h}{P} \sum_{\substack{1 \leq p \leq P+1 \\ 1 \leq p+h \leq P+1}} H_I(A_a^p x_a, x_{a'}, A_c^{p+h} x_c, x_d).$$

The sum over p stays inside the cell contribution. When the current group is also summed, its sum stays there as well. This is the expression to which the short-block estimate is applied; taking absolute values of its individual bands would lose the matched-band summation used in that estimate.

Forward only the transferred main term in (4.16) to the next step. Removing the remaining positive filters in decreasing rank ends at

$$(4.18) \quad H_I(\mathcal{D}_a S_a, S_{a'}, B_c, B_d), \quad \mathcal{D}_a \text{ a product of } O(\log \mathcal{L}) \text{ helper averages.}$$

Restoring B_c, B_d to the full Y_c, Y_d uses their pair bound from the base reduction. Each correction has an additional basic U factor and is therefore a three-side term.

For the retained defects, the endpoint convention matters. If the outgoing filter is a ball filter, split $A_c^{P+1} = 1 - Q_{c,P}$ into the shared identity and its positive ball endpoint. The other complementary band endpoints have usable ball ranges. If the outgoing filter is cumulative, its outer ball has already been removed and the entire complementary band expression is shared across groups. Role d likewise either has one positive outer ball filter or is shared. In the endpoint energy estimates, a shared identity is evaluated on the union of required cells and counted once; the original form still retains every actual use. These are exactly the endpoint classes controlled by (4.14) and [V, Corollary 6.2]; they give the side estimate in Section 5 its coupled band operands.

We have obtained the base minimum, the localized group expressions, their outgoing-filter identity, and the endpoint estimates with the same main-column budgets as H and V. The remaining work is to estimate the coupled near bands and the remaining localized quadratic contributions. Section 5 establishes the coefficient-dependent comparisons used for those two estimates.

5. SHORT BLOCKS, RATIONAL SHIFTS, AND THE CORE ESTIMATE

Starting from the localization proved in the preceding section, we now split the arguments into side terms and core terms. Side terms are controlled by square estimates for projection increments, together with a comparison on a bounded block of depths. Core terms are controlled by separation of the frequencies left after removing a common quadratic phase. We prove the

coefficient-dependent parts of both arguments here. The projection, graph, and amplitude summations that we use from [H, V] are identified at their points of application.

Throughout this section, the primitive integer vector $b = (0, b_1, b_2, b_3)$, the integers e_j , and the nonzero integers c_j are those of Lemma 2.3. Constants may depend on these fixed integers and on the admissible kernel bounds. We write $e(t) = \exp(2\pi it)$. For an interval I , the notation $\|f\|_{2,I}$ denotes $(|I|^{-1} \int_I |f|^2)^{1/2}$, also for Hilbert-space-valued functions.

5.1. A comparison on a short block. We first establish the version of [H, Lemma 7.2] needed by the side argument. This estimate compares randomized inputs with the fixed-input constant $C_N(q)$; in particular, the exponent of the block length must tend to zero as q tends to two.

Proposition 5.1 (Short-block comparison). *Let $2 < q < 3$. Let a finite dyadic tree piece have root R , with internal summation intervals in at most K consecutive depths and within the original range of at most N depths. Its terminal intervals, called leaves, partition R and occur in this depth block or one depth below it. In role j , let $\mathbf{z}_j$ be a deterministic Hilbert-space-valued function supported on R , with $\|\mathbf{z}_j\|_{2,L} \leq s_j$ on every leaf L . Suppose scalar randomizations z_j^σ satisfy*

$$(\mathbb{E}_\sigma |z_j^\sigma(y)|^a)^{1/a} \leq C_a \|\mathbf{z}_j(y)\|$$

for $a = 2, 4, 8$, with constants independent of the tree and K . For measurable signs $|\epsilon_I(\sigma)| \leq 1$ and any subcollection of internal intervals,

$$\mathbb{E}_\sigma \left| \sum_I |I| \epsilon_I(\sigma) H_I(z_0^\sigma, z_1^\sigma, z_2^\sigma, z_3^\sigma) \right| \leq (1 + C_N(q)) C_{q,b} (1 + \log(K+2))^C (K+2)^{11(q-2)/(q-1)} |R| \prod_j s_j.$$

The constant $C_{q,b}$ may also depend on the fixed moment bounds C_a and the kernel bounds.

Proof. Homogeneity reduces to $s_j = 1$; a zero size makes the form zero. Multiplying all signs by one random scalar of modulus one reduces the expected absolute value to the absolute value of an expected signed sum.

We use the partition of the four leaf depths from the proof of [H, Lemma 7.2]. Here is the information from that partition that will be needed. If the least depth occurs more than once, designate two roles attaining it. If it occurs once, designate that role and separate the least and second least depths by the smallest dyadic interval of depth indices containing them in opposite halves. Specify the nonempty set of roles whose depths belong to its upper half, and designate one of those roles by a fixed ordering. Every role that is neither the minimum role nor in that specified set has depth greater than the upper endpoint of that dyadic interval. This partitions the depth tuples into rectangles of masks E_j , each a union of leaves. There are $O(1 + \log(K+2))$ length classes and a bounded number of role assignments. Within each class, there are $O(K+1)$ positions; the masks in each of the two designated roles, say a, a' , are disjoint over those positions. For a fixed rectangle there is an index t such that every role has leaf depth at least t , with the possible exception of role a . Only summation depths $s < t$ contribute. Indeed a leaf meeting an internal interval must be its strict descendant. The same assertion holds after the small coordinate enlargements below, by the kernel's fixed interior support margin.

Write r_s for the length at depth s , and set $\delta = \beta r_t$, where $\beta > 0$ is a sufficiently small fixed dyadic number depending on b and the support margin. Decompose the physical coordinates as

$$x = \delta(n + u), \quad \tau = \delta(m + v), \quad n_j = n + b_j m, \quad w_j = u + b_j v.$$

The slow coordinates n, m are integers. Choose the fast representatives from the parallelogram

$$(5.1) \quad \mathcal{P}_b = \{(u, v) : 0 \leq u < 1, \quad 0 \leq u + b_1 v < |b_1|\}.$$

This is a fundamental domain for translation by $\mathbb{Z}^2$, of area one. To decompose a point, first choose n so that its first coordinate minus n belongs to $[0, 1)$, and then choose m so that $u + b_1 v$ belongs to $[0, |b_1|)$. The latter choices differ by b_1 , so existence and uniqueness hold for either sign of b_1 .

Let $\mathbb{T}_0 = \mathbb{R}/(T_0\mathbb{Z})$, with Haar probability measure, where T_0 will be a sufficiently large fixed number. Define $F_j^\sigma(p, w)$ by sampling $z_j^\sigma(\delta(p + w))\mathbf{1}_{E_j}(\delta(p + w))$ on a bounded interval of real lifts and setting it to zero elsewhere on $\mathbb{T}_0$. Use lift intervals $[0, 1)$ and $[0, |b_1|)$ in roles zero and one, and bounded intervals containing the ranges of w_2, w_3 on $\mathcal{P}_b$ in the other roles.

The fast Haar integral reproduces integration over $\mathcal{P}_b$, up to its fixed normalization. To check the potentially noninjective pair coordinates, suppose a torus progression has all four bounded lifts $\tilde{w}_j$ in these windows. Put

$$\tilde{u} = \tilde{w}_0, \quad \tilde{v} = \sum_j e_j \tilde{w}_j.$$

The integer inverse of Lemma 2.3 gives $\tilde{w}_j = \tilde{u} + b_j \tilde{v} \pmod{T_0}$. All the discrepancies are bounded in terms of the fixed windows and b, e . Choose T_0 larger than these bounds. Each discrepancy then vanishes, so these are equalities of real numbers and $(\tilde{u}, \tilde{v}) \in \mathcal{P}_b$. Conversely, every point of the tile gives exactly these lifts. This verifies the sampling identity without asserting that any coordinate pair is unimodular.

Replace the kernel by its value at $(\delta n, \delta m)$. At depth s , the error is $O_b(\delta/r_s)$, with a fixed support enlargement. The normalized sampled form has factor $(\delta/r_s)^2$. On the product of local counting measure and the fast torus, any two distinct coordinate maps are injective on the lattice and Haar-surjective on the torus. Consequently the two- L^1 , two- L^∞ bounds hold, and interpolation gives the absolute form estimate whenever the four reciprocal exponents sum to at most two.

For clarity, let $M_{j,I}$ be the squared mass of $\mathbf{z}_j$ in all masked sampling windows relevant to I , including their bounded overlap multiplicity. Choosing β small enough puts these windows inside I . The leaf normalization therefore gives

$$M_{j,I} \leq C_b |I|, \quad \sum_{\text{depth}(I)=s} M_{j,I} \leq C_b |E_j|.$$

The absolute L^2 estimate and the moment hypothesis bound the sampling errors at depth s by

$$C_b \frac{\delta}{r_s} |E_a|^{1/2} |E_{a'}|^{1/2}.$$

The sum over $s < t$ is bounded, and summing rectangle positions uses the disjointness of both designated masks.

We next improve the integrability of the fast inputs. On their fast Fourier coefficients define

$$\widehat{U_{j,\gamma} F}(p, k) = e(\gamma k^2 / c_j) \widehat{F}(p, k), \quad \gamma \in \mathbb{R}/(C_c \mathbb{Z}), \quad C_c = \text{lcm}_j |c_j|.$$

The fast progression integral is unchanged under simultaneous application of these unitaries. Its Fourier support obeys $\sum k_j = \sum b_j k_j = 0$, so Lemma 2.3 gives $k_j = c_j(h + b_j h')$ for real h, h' . Thus

$$\sum_j k_j^2 / c_j = \sum_j c_j (h + b_j h')^2 = 0.$$

For a single role, integrating the fourth power first in the fast variable and then in γ equates the sums and the sums of squares of two pairs of Fourier indices. Those pairs agree as multisets, whence

$$(5.2) \quad \mathbb{E}_\gamma \int_{\mathbb{T}_0} |U_{j,\gamma} F(p, w)|^4 dw \leq 2 \|F(p, \cdot)\|_2^4.$$

Fourier polynomial approximation is justified by the pairwise L^2 form estimate and this same fourth-moment estimate for differences.

Let $\mathfrak{e}_j(p)$ now denote the fast L^2 size of the deterministic stack in the sampling window. Window overlap gives $\delta \sum_p \mathfrak{e}_j(p)^2 \leq C_b |E_j|$. If all leaves of a role have depth at least t , then $\mathfrak{e}_j(p) \leq C_b$: disjoint leaves meeting a window of length $O_b(\delta)$ have total length at most $O_b(\delta) + 2r_t$. In every role, including the possible remaining role a ,

$$\frac{\delta}{|I|} \sum_{p \text{ relevant to } I} \mathfrak{e}_j(p)^2 \leq C_b.$$

Minkowski's inequality transfers the assumed sign moments to these window sizes. Intermediate moment orders below eight are bounded by the eighth moment using Hölder's inequality.

Write $G_j = U_{j,\gamma} F_j$. If role a has leaves above depth t , clip it radially at $A = (K+2)^{10}$. The clipped tail has local L^1 norm at most $A^{-1} \|F_a\|_2^2$, whose L^4 norm in the random parameters is $O_b(A^{-1})$. In the other roles the local L^3 norms have bounded fourth moments, by (5.2) and the preceding window bounds. The absolute form estimate with exponents $(1, 3, 3, 3)$ therefore gives expected tail error $O_b(A^{-1})$ per normalized cell. Even summing $O((K+1)^2)$ positions and depths is harmless. For the rest of the proof, replace G_a by this clipped function when that role is present.

To return to the original continuous form, fix the random parameters and $(u, v) \in \mathbb{T}_0^2$, and plant bursts

$$g_j(y) = \sum_p G_j(p, u + b_j v) \mathbf{1}_{\{|y - \delta p| < c' \delta\}}.$$

Choose $c' > 0$ so small that $c'(2 + \max_j |b_j| \sum_l |e_l|) < 1$. If a real configuration lies in bursts centered at p_j , set $n = p_0$, $m = \sum_j e_j p_j$. Then

$$|x/\delta - n| < c', \quad |\tau/\delta - m| < c' \sum_j |e_j|, \quad |p_j - n - b_j m| < 1.$$

The last quantity is an integer and hence is zero. For this slow pair the allowed offsets form the polygon $\{|u' + b_j v'| < c' \ (0 \leq j \leq 3)\}$, of fixed positive finite area. The center-valued kernel integral on the polygon therefore gives exactly the sampled product, times that area and δ^2 . The second kernel replacement has the same error estimate as the first: unitarity and clipping preserve fast L^2 bounds and do not enlarge the set of slow indices. We can now apply $C_N(q)$ to these fixed compactly supported burst inputs, using the original admissible kernels.

For completeness, the exponent calculation is unchanged. Put

$$p_a = p_{a'} = p_* = 2 + 2(q-2), \quad p_j = \frac{2p_*}{p_* - 2} \quad (j \notin \{a, a'\}).$$

Use fast exponent p_* in the clipped role and exponent four in the other roles. Fast pairwise Hölder, random Hölder of exponent four, spatial Hölder with the p_j , and the mixed-norm maximal theorem [H, Lemma 7.1] apply: all exponents in that maximal theorem exceed q . The resulting regular-role factors are $C_b |E_j|^{1/p_j}$; the clipped factor is $C_{q,b} A^{1-2/p_*} |E_a|^{1/p_*}$. Summing $O(K+1)$ positions in the two disjoint designated masks costs $(K+2)^{1-2/p_*}$. Hence the total power is

$$A^{1-2/p_*} (K+2)^{1-2/p_*} = (K+2)^{11(q-2)/(q-1)}.$$

The logarithmic number of length classes finishes the proof. $\square$

This proposition supplies exactly the input used by the side estimates [H, Lemmas 7.3–7.4 and Proposition 7.5] and [V, Section 6.5]. Their remaining arguments concern nested projections, frozen stacks, and variation in depth. In addition to the local absolute form bound, they assume

the square estimate $H(17)$ for the linear sides. We identify its source below; no coordinate substitution is needed in those projection arguments.

5.2. A common gauge and a lattice of rational shifts. We describe the data from the mode construction of [H, Section 8] before checking its arithmetic. Fix a localized group G , with central tag $z = (\theta_z, C_z, \lambda_z)$, and a role l . A *primary* is an available main column of this role's outer projection. Its first matching cell is the cell where the finite expansion described below is chosen; that choice and its labels are then inherited on descendants. Thus later freezing does not resample a primary's expansion.

Choose an integer Q_0 clearing the denominators of C_z , and put $\vartheta = \theta_z/Q_0$. For an offset ρ in the unit cube of horizontal coordinates, define

$$d^\rho(y) = \rho + \{\theta_z y - \rho\}, \quad p^\rho(y) = \lambda_z y^2 + (\theta_z y)^T C_z d^\rho(y).$$

Choose $\delta_c = \exp(-k^{C_{\text{cut}}})$ sufficiently small after the main budgets, with fixed exponent C_{cut} . Let ζ^ρ be the product Gevrey cutoff of [H, Section 8], vanishing within distance δ_c of each branch cut and equal to one outside distance $2\delta_c$. It is bounded between zero and one. For each role and group, one offset is kept fixed throughout its history. Its expectation is at least $1/2$, and $\mathbb{E}_\rho |1 - (\zeta^\rho)^2|^2 \leq C d_{\text{max}} \delta_c$, where d_{max} denotes an upper bound for the dimensions.

For a primary $\Phi = (\varphi_\Phi, \epsilon_\Phi^{1/2} \mathbf{e}_\Phi)$ with normalized tag v , the endpoint splitting from the localization section and the chart expansion of [H, Lemma 4.4] give, at its first match,

$$\zeta^\rho \varphi_\Phi e(-c_l p^\rho) = \sum_m a_{\Phi, m} e(\lambda_m y^2 + \gamma_m y) + O(\delta_c).$$

The chart expansion uses this same error tolerance δ_c , and all modes of one primary have the same quadratic coefficient. As in [H, Lemma 8.1], use mode atoms

$$\psi_m(y) = \left(\zeta^\rho(y) e(c_l p^\rho(y)) e(\lambda_m y^2 + \gamma_m y), \epsilon_m^{1/2} \mathbf{e}_m \right).$$

The coefficient sum, mode count, normalized curvature, and rational heights have logarithms polynomial in the accuracy and localization parameters k, P . In particular, for modes of one primary,

$$(5.3) \quad \gamma_m - \gamma_{m'} = a \cdot (\theta_v, \vartheta), \quad a \in \mathbb{Q}^{\dim(\theta_v, \vartheta)} \text{ of controlled height.}$$

Multiplication of the splitting by the fixed integer c_l changes only fixed constants in these bounds.

The resulting lift $\mathcal{R}_{l, G}$ is one fixed pointwise bounded operator on augmented inputs. Indeed, set $q_\Phi = \sum_m a_{\Phi, m} (\psi_m)_{\text{phys}} - (\zeta^\rho)^2 \varphi_\Phi$, so that $|q_\Phi| \leq \delta_c$. For an augmented input F , its lifted physical component is

$$(\zeta^\rho)^2 F_{\text{phys}} + \sum_{\Phi \text{ matched}} \epsilon_\Phi^{-1/2} F_\Phi q_\Phi,$$

and its mode coordinate belonging to (Φ, m) is $\epsilon_m^{1/2} a_{\Phi, m} \epsilon_\Phi^{-1/2} F_\Phi$. Choose δ_c small and ϵ_m small after the main column budgets. Cauchy–Schwarz then gives a uniform operator norm. The private coordinate of an outer projection output records its column coefficient, so this operator gives exactly its mode synthesis. It is cell-local and independent of the freezing block.

There is also a common periodic multiplier for the four gauges. For four offsets, put $d_j = d^{\rho_j}(y_j)$ and

$$d_\diamond = \sum_j e_j d_j, \quad k_j = d_j - d_0 - b_j d_\diamond.$$

The k_j are bounded integer vectors. Between branch cuts the lifts are affine and their affine parts cancel in this expression; hence the carries are locally constant. The three identities $\sum c_j b_j^a = 0$, $0 \leq a \leq 2$, give

$$(5.4) \quad \prod_j e(c_j p^{\rho_j}(y_j)) = e \left(\sum_j c_j (\theta_z y_j)^T C_z k_j \right).$$

For fixed carries the right side is an ordinary character on the torus with coordinates $(\vartheta y_{l'}, \vartheta t)$, for any anchor l' : Q_0 clears C_z , and changing the anchor uses integer coefficients $b_j - b_{l'}$. Smooth cutoffs slightly larger than the original ones extend these character pieces by zero at the cuts. They give the smooth periodic multiplier, bounded by one, required in [H, Section 8.3].

The number of carries and the derivative bounds acquire only fixed factors per coordinate or derivative. Their logarithmic costs remain polynomial in P, k . Choose a mode budget $D_* \geq e^k$, $\log D_* \leq \text{poly}_b(P, k)$, covering the mode lists, coefficient sums, heights, inverse private weights, and absolute Fourier sums, and large enough to dominate the dimensions and all fixed slope factors. Only then choose a Fourier cutoff B_* with tails at most D_*^{-300} . This preserves the order of choices in H; a coefficient-sum budget is not defined through its own truncation radius.

To control rational divisions at later frequency tests, fix an integer

$$(5.5) \quad D_s \geq 6, \quad 6 \mid D_s, \quad |b_i - b_j| \mid D_s \quad (i \neq j).$$

The mode graphs will use shifts in the lattice generated by ϑ/L_h , where

$$(5.6) \quad L_h = D_s^h, \quad E_h = A_0^{h+1}.$$

Here E_h bounds integer shift indices. The choice of D_s is fixed independently of all depth and accuracy parameters. From this point, constants indexed by b may also depend on this fixed choice of D_s .

We recall the scale geometry to make the graph conditions explicit. In a depth block $[s_0, s_0 + K)$, freeze coefficients at $u_{s_0} = s_0 - 4K$, compare on a cell J at depth $v_{s_0} = s_0 - 2K$, and put $t_{s_0} = s_0 + K$. A mode has instantaneous frequency $\omega_m(x) = 2\lambda_m x + \gamma_m$. On the list frozen at u_{s_0} , join two modes in the PRE graph if

$$\sup_{x \in J} |\omega_m(x) - \omega_{m'}(x) + n \cdot \vartheta/L_h| \leq \Xi/r_{v_{s_0}} \quad \text{for some } n \in \mathbb{Z}^{\dim \theta_z}, \quad |n|_\infty \leq E_h.$$

The CURRENT graph has the same definition with threshold $\Xi/r_{t_{s_0}}$. Write $\mathcal{E}_h(m)$, $\mathcal{F}_h(m)$ for their components. Both graphs increase with height, because a witness n at height h becomes $D_s n$ at height $h + 1$, provided $A_0 \geq D_s$. They also increase along successive depth blocks on inherited lists.

Here are parameter choices sufficient for the arithmetic. Let $d_{\max} \geq 2$ bound all relevant dimensions, set $\Delta = 10$, and put

$$C_* = (10d_{\max} D_*)^{100d_{\max}^4}, \quad R' = \lceil \log_2 C_* \rceil + 2\Delta + 10.$$

Choose the largest height J_* , with $0 \leq h \leq J_*$, large compared with $\delta_*^{-2} d_{\max} R' \log(2D_m)$, where D_m is the main-column budget and $\delta_* > 0$ is a sufficiently small fixed number. Take A_0 to be a sufficiently large fixed power of $C_* D_s^{R'} B_* D_*$, and $\Xi = D_*^{1000}$. Choose a rational height reserve and a frequency gap with

$$\mathcal{H} > (d_{\max} C_* E_{J_*} B_* L_{J_*} Q_0 \Xi)^{10}, \quad \mathcal{T} > (d_{\max} \mathcal{H})^{C d_{\max}^4 \Xi}.$$

Finally choose K , polynomial in P, k , after these logarithmic budgets, so the height-bin packets of [H, Lemma 3.5] and the gap $\mathcal{T}$ fit inside a block. Increase it so the pre-lag Taylor errors are as small as any prescribed fixed inverse power of D_* . All degrees can be fixed independently of q .

For a primary with horizontal parameters $\Theta = (\theta_v, \vartheta)$, call a block nonexceptional when every rational vector a of height at most $\mathcal{H}$ satisfies

$$(5.7) \quad |a \cdot \Theta| \leq \mathcal{T}^{-1}/r_{v_{s_0}} \quad \text{or} \quad |a \cdot \Theta| \geq \mathcal{T}/r_{t_{s_0}}.$$

The height-bin proof of [H, Lemma 8.2] gives $O(d_{\max})$ exceptional blocks per primary. At a nonexceptional block let K_{rat} be the rational span of the tiny vectors, and set $K_B = \{w : (0, w) \in K_{\text{rat}}\}$. Cramer's rule and the choice of $\mathcal{T}$ show that every height- $\mathcal{H}$ vector in K_{rat} is itself tiny.

Lemma 5.2 (Height stabilization). *For a nonexceptional primary, all but $O(d_{\max} R')$ heights $h \in \{0, \dots, J_*\}$ have the following property: two of its modes in one CURRENT component at height h are directly adjacent in PRE at height $h - \Delta$.*

Proof. This is [H, Lemma 8.3] with the shift lattice (5.6); we give its divisibility and size argument. A simple CURRENT path has at most D_* edges. The equal quadratic coefficients of its endpoint modes and (5.3) yield

$$a + (0, m_h/L_h) \in K_{\text{rat}}, \quad m_h \in \mathbb{Z}^{\dim \theta_z}, \quad |m_h|_\infty \leq D_* E_h.$$

Indeed the path bounds its evaluation by $D_* \Xi/r_{t_{s_0}}$, and the gap forces the tiny alternative.

Let V_i be the rational span of the height- D_* vectors q for which $q + (0, m/L_i) \in K_{\text{rat}}$ with $|m|_\infty \leq D_* E_i$. Let U_i be the span of the integer vectors in K_B of size at most $C_*^3 D_* E_i$. These spaces increase. Discard heights below R' and those where either rank changes between $h - R'$ and h ; there are $O(d_{\max} R')$ such heights.

At a remaining height, express $a \in V_{h-R'}$ in a defining basis $a = \sum_j \beta_j a_j$, with witnesses m_j . Cramer's rule bounds the coefficients and their common denominator by C_* . Subtracting the relations gives

$$m_h - D_s^{R'} \sum_j \beta_j m_j \in K_B.$$

After clearing denominators its size is at most $C_*^3 D_* E_h$, so it lies in $U_h = U_{h-R'} =: U$. The quotient $\mathbb{Z}^{\dim \theta_z} / (U \cap \mathbb{Z}^{\dim \theta_z})$ is free, since the intersection with the rational space U is saturated. In an integer basis of this quotient,

$$[m_h] = D_s^{R'} \sum_j \beta_j [m_j].$$

For each prime $p \mid D_s$, every integral coordinate on the right has valuation at least $R' v_p(D_s) - \log_2 C_* \geq \Delta v_p(D_s)$. Thus $[m_h]$ is divisible by D_s^Δ .

Choose an integral lift m' of this quotient class divided by D_s^Δ . Then

$$m' - D_s^{R'-\Delta} \sum_j \beta_j m_j \in U.$$

Round its coefficients in an integer spanning set for U whose vectors have size at most $C_*^3 D_* E_{h-R'}$. The adjusted lift remains integral and satisfies

$$|m'|_\infty \leq D_s^{R'-\Delta} \sum_j |\beta_j| |m_j|_\infty + d_{\max} C_*^3 D_* E_{h-R'} \leq E_{h-\Delta}.$$

The last inequality follows from the stated fixed-power choice of A_0 . No lattice basis or covolume estimate is involved. Since $m_h - D_s^\Delta m' \in K_B$, the vector $a + (0, m'/L_{h-\Delta})$ belongs to K_{rat} and remains within the height reserve. It is tiny by (5.7), which gives the required PRE edge. $\square$

We now specify the remaining output imported from [H, Section 8]. Write $\pi(\mathcal{D})$ for the primary labels represented in a set $\mathcal{D}$ of modes. A mode is good at height $h \geq \Delta$ when

$$\lfloor \log_{1.1} |\pi(\mathcal{E}_{h-\Delta}(m))| \rfloor = \lfloor \log_{1.1} |\pi(\mathcal{F}_h(m))| \rfloor$$

and at least nine tenths of the primaries in $\mathcal{F}_h(m)$ are nonexceptional and satisfy Lemma 5.2 at h . Heights below Δ are bad. Call a mode core if it is bad at at most $\delta_*(J_* + 1)$ heights; call every other mode a linear-side mode. The increasing graphs and height stabilization give the coherence of these predicates, the bad-density count, and hereditary representatives by exactly [H, Lemmas 8.4–8.8, Proposition 8.9, and Lemma 8.10]. Those arguments use component inclusion, cardinality, projections, and the rational gap; their coordinate inputs have all been established above. A minimal PRE or CURRENT component means a component at height zero. Good modes in a CURRENT component have one PRE predecessor at height $h - \Delta$. Moreover, core modes in one minimal CURRENT component have identical good-height predicates throughout the upper range

$$\mathcal{H}_{\text{up}} = \{h \in \mathbb{Z} : \lceil 4\delta_* J_* \rceil \leq h \leq J_*\}.$$

Each core mode has bad fraction at most $2\delta_*$ in this range.

We also record the decomposition of the original localized outputs. Let $Z_{l,G,s}$ be the current augmented output in role l . For a basic- B role in the block beginning at s_0 , put

$$\tilde{Z}_{l,G,s_0} = \mathcal{R}_{l,G} Z_{l,G,u_{s_0}}.$$

Let A_{l,G,s_0} and V_{l,G,s_0} be its exact core and linear-side mode subtotals, with all private coordinates retained. Define the full side by

$$S_{l,G,s} = Z_{l,G,s} - Z_{l,G,u_{s_0}} + V_{l,G,s_0} \quad (s_0 \leq s < s_0 + K).$$

In a basic- U role set $S_{l,G,s} = Z_{l,G,s}$ and $A_{l,G,s_0} = \mathcal{E}_{l,G,s_0} = 0$. On physical components these definitions give

$$Z_{l,G,s} = A_{l,G,s_0} + S_{l,G,s} + \mathcal{E}_{l,G,s_0}.$$

For a basic- B role, the physical error is

$$\mathcal{E}_{l,G,s_0} = (Z_{l,G,u_{s_0}})_{\text{phys}} - (\tilde{Z}_{l,G,s_0})_{\text{phys}}.$$

The offsets can be fixed so that every term containing this physical error is negligible. The explicit lift, applied to $Z_{l,G,u_{s_0}} = \sum_{\Phi} \beta_{\Phi} \Phi$, gives

$$\mathcal{E}_{l,G,s_0} = (1 - (\zeta^{\rho})^2)(Z_{l,G,u_{s_0}})_{\text{phys}} - \sum_{\Phi} \beta_{\Phi} q_{\Phi}.$$

The main coefficients and original actual-use cells were fixed before the offsets. Their coefficient, size, and path-count bounds, together with the cutoff expectation, imply the error estimate from [H, Section 8, subsection on fixing offsets]:

$$\mathbb{E}_{\rho} \sum_{l,G,I \text{ used}} |I| \|\mathcal{E}_{l,G,s_0(I)}\|_{2,I}^2 \leq D_m^C d_{\max} \delta_c |R|.$$

With δ_c chosen this small in the main budget, fix offsets satisfying this bound. All later selections only remove nonnegative terms; they need not be independent of the offsets. The linear-side square estimate is uniform in the offsets. The local absolute L^2 form bound and Cauchy–Schwarz therefore make the terms containing $\mathcal{E}$ negligible, after the same size stops as in H. This explains why the subsequent estimates may use the exact core and full-side arguments.

Here are the quantitative conclusions used below. A successful CURRENT component contains a good core mode, whose PRE component at $h - \Delta$ is its uniquely determined good predecessor. For fixed height, role, group, and the pair of floor logarithms of the predecessor's primary and mode counts, choose a representative mode inherited under predecessor inclusion. On comparable actual use cells I, I' , distinct representatives satisfy H(25), namely

$$(5.8) \quad |\omega_a(x_*) - \omega_{a'}(x_*) + n \cdot \vartheta / L_h| > \frac{\Xi}{2} \max(|I|^{-1}, |I'|^{-1}), \quad |n|_{\infty} \leq E_h,$$

for $x_* \in I \cap I'$. On an actual cell, a successful component $\mathcal{F}$ has amplitude

$$\sigma_{\mathcal{F}}^2 = \sum_{\mathcal{D} \subseteq \mathcal{F}} \left(\|A_{\mathcal{D}}\|_{2,I} + D_*^{-30} \sum_{\substack{m \in \mathcal{D} \\ m \text{ core}}} |\alpha_m| \right)^2.$$

Here $\mathcal{D}$ runs over minimal CURRENT components, $A_{\mathcal{D}}$ is the full core subtotal in the augmented mode space, and α_m is its frozen mode coefficient. This is H(26); the added coefficient terms absorb uniform Taylor and Fourier errors.

The Bessel estimates and bad-density count give H(17):

$$\sum_{G,s_0} \|V_{l,G,s_0}\|_2^2 \leq G_0 |R|, \quad G_0 = \exp(C_{q,b}(1 + \log(2k))^C).$$

This is the square hypothesis needed for the side estimates following Proposition 5.1. After the stopping construction of [H, Proposition 8.12], the representatives occurring in a dyadic amplitude bin σ_j along a retained path satisfy H(28):

$$(5.9) \quad \sum_G |\mathcal{A}_j^G| \leq \min(D_*, P^C \sigma_j^{-2}), \quad \sum_{s,G} \|S_{l,G,s}\|_{2,I_s}^2 \leq P^C.$$

Here $\mathcal{A}_j^G$ is the set of distinct representative labels in that bin. All local side and component sizes are at most P^C .

For scheduled inputs, these statements are used in the form proved in [V, Section 6.3–6.4, especially Lemma 6.3]. The rows are cell-local projections composed with the fixed bounded lift $\mathcal{R}_{l,G}$. Their ordinary Bessel estimates come from the separated mode construction, and their adjoint main ranges lie in the usable endpoint spaces established by localization. V evaluates the depth input after the final main basic projection. Its row composition result therefore applies to the same size, variation, reading-depth, and endpoint hypotheses. In particular, the possibly large mode budget D_* is not inserted into the strong square constant G_0 . This is the distinction required by V's side estimate and by the final choice of q .

5.3. The side estimate and the near-band gain. We now estimate the contribution of the full sides to the cell minimum in (4.7). The input is the localized expression (4.13), its outgoing-filter identity, the endpoint variation estimate (4.14), and the linear-side square bound H(17). These are precisely the hypotheses of [H, Proposition 7.5]; for scheduled inputs their counterparts are assembled in [V, Section 6.5].

Expand each localized argument into its core, full side, and negligible physical error. Terms with at least three side factors have a length-weighted p -power bound $P^C |R|$, for any fixed $2/3 < p < 1$. Indeed Cauchy–Schwarz first sums the group index in two side roles; writing the three resulting square sizes as a_s, b_s, c_s , their contribution is bounded using

$$\sum_s (a_s b_s c_s)^p \leq \left(\sum_s a_s^2 \right)^{p/2} \left(\sum_s b_s^2 \right)^{p/2} \left(\sum_s c_s^2 \right)^{p/2}.$$

This uses $3p > 2$ and the retained path caps; the fourth argument uses its P^C size bound. The group sum is formed before the power is taken.

In a term with exactly two sides, restore its two core complements to the whole current arguments $Z_{c,G,s}, Z_{d,G,s}$. The corrections have a third side. The outgoing-filter identity then removes the complementary projections one at a time, transferring helper averages to one side, say role a . Its main term is

$$H_I(\mathcal{D}_{a,s} S_{a,G,s}, S_{a',G,s}, B_c, B_d),$$

where $\mathcal{D}_{a,s}$ is the resulting product of helper averages. Restoring the full base pair Y_c, Y_d introduces another third-side error. The pair-smallness from the base reduction bounds this main term by η times the two side sizes. Summing costs at most $\eta P^C |R|$.

At one transfer step, use the outgoing and helper bands defined in (4.15), with these two full sides as its fixed side operands. The exact outgoing-minus-helper defect is (4.16). Its far bands have negligible error. For each remaining offset h , $|h| \leq 2$, the retained expression (4.17) has coefficient $-h/P = O(P^{-1})$. Its paired sum over the band index must remain inside the cell contribution.

Freeze the complementary band stack and the other complementary argument at the block's first depth. Their differences from current values have square budgets by (4.14). Freeze errors therefore supply a third strong factor and have the preceding p -power bound. The same argument handles cells where a frozen complement has large size: its difference from its bounded current value is then large. On the remaining ancestor hulls inside a block, both frozen complements have bounded local L^2 size.

The binary-prefix decomposition in [H, Lemma 7.4] now writes the two varying sides, at their uses, as selections from fixed vector stacks. The terminal complementary band $1 - Q_{c,P}$ is first split into the identity and a positive ball term. Positive outer-ball ranges have joint group budgets, whereas group-independent identity and cumulative histories are counted once rather than once per group. This convention is needed for the following stack energies. One stack includes the band index. There are only G_0 overhead factors. Individual pad choices stay inside the actual positive averaged operators: a choice is coupled across depths, and Jensen's inequality restores the average after estimating it. They are not additional stack configurations. The stack energies satisfy H(18):

$$\sum_{\nu} \|\mathbf{A}_{\nu}\|_2^2 \leq PG_0 |R|, \quad \sum_{\nu} \|\mathbf{B}_{\nu}\|_2^2 \leq G_0 |R|.$$

Here ν records the group, block, and fixed stack configuration. The functions remain fixed after subsequent cuts. Independent Rademacher signs select the two position indices, and a third family matches the band in $\mathbf{A}_{\nu}$ with the frozen complementary band stack. Iterated Khintchine gives the moment hypothesis of Proposition 5.1.

For precision, the passage from these energies to a sum of local stack sizes uses the common hull cuts of [H, Section 7]. A hull consists of all ancestors between a use and its maximal use cell, including their full intervals. Write Ω_{ν} for the union of the original use cells. The main path and catalog counts give $\sum_{\nu} |\Omega_{\nu}| \leq M_0 |R|$, with $\log M_0 \leq \text{poly}_{q,b}(k)$. For a family of total stack energy at most $E |R|$, write $\tilde{m}_{\nu}$ for its full hull maxima. They obey

$$\sum_{\nu} |\{x : \tilde{m}_{\nu}(x) > T\}| \leq \min(M_0 |R|, CE |R|/T).$$

Stopping at level $C_1 E$ and integrating up to that level costs at most $CE(1 + \log(2 + C_1 M_0)) |R|$. Thus only a main-budget logarithm enters G_0 . For the retained hull define

$$m_{A,\nu}(x) = \sup_{\substack{I \text{ in the retained hull} \\ x \in I}} \frac{1}{|I|} \int_I \|\mathbf{A}_{\nu}(y)\|^2 dy,$$

with value zero for an empty supremum; define $m_{B,\nu}$ in the same way. The weak L^1 estimate for the dyadic maximal function, followed by truncation and integration, gives common individual cuts; Markov's inequality for the truncated sum gives the joint cuts for which these squared hull maxima satisfy

$$\sum_{\nu} m_{A,\nu} \leq PG_0, \quad \sum_{\nu} m_{B,\nu} \leq G_0.$$

The removed roots have arbitrarily small fixed total relative length. They are first crossings of partial ancestor quantities, so retained caps hold on whole intervals even in branches containing a later cut. A sparse decomposition of each retained hull into tree pieces gives disjoint major subsets of the piece roots. If a_Q, b_Q are the two normalized stack sizes at such a root, then

$$\sum_{\nu, Q} |Q| a_Q b_Q \leq C \int_R \sum_{\nu} \sqrt{m_{A,\nu} m_{B,\nu}} \leq P^{1/2} G_0 |R|.$$

Complete each piece by its boundary children. The two side stacks then have the required leaf-size bounds, and the frozen complements have bounded leaf sizes. Proposition 5.1 applies to at most K depths of each piece. Multiplying by the transfer coefficient $O(P^{-1})$ yields the near-band bound

$$(5.10) \quad (1 + C_N(q)) C_{q,b} P^{-1/2} G_0 (K + 2)^{C_{19}(q-2)} |R|.$$

The logarithmic factor of the short-block estimate is included in G_0 . Arbitrary signs at the original cells are allowed, so this is an absolute estimate for their coupled band contributions. No absolute sum over individual paired band indices has been used.

Thus [H, Proposition 7.5], with the coefficient-dependent comparison now proved, gives a complete conditional side bound. Third-side and freeze errors have p -power cost $P^C |R|$; transferred main terms have absolute cost $\eta P^C |R|$; the remaining near bands satisfy (5.10). Applying $\min(\eta, x) \leq \eta^{1-p} x^p$ only to the powered terms gives their contribution $C_{q,b} \eta^\varepsilon P^C |R|$, with $\varepsilon = 1 - p > 0$. All cuts remove original summands at first crossings and the data are reconstructed on each normalized subroot. This is the root-uniform side estimate used in the closing argument. The scheduled version follows from the same fixed stacks and the row estimates already stated, as in [V, Section 6.5].

5.4. Frequency tests for three or four core arguments. Fix a retained path point x_* , an actual cell $I = I_s$ with $r = |I|$, a group, and a common successful height $h \in \mathcal{H}_{\text{up}}$ for the core arguments. Put $R_s = r^{-1}$ and

$$\xi = \vartheta / L_{h-\Delta}, \quad \gamma_j^* = \omega_{a_j}(x_*), \quad B'_h = D_*^5 E_{h-\Delta}.$$

The representative a_j belongs to the good predecessor of the component in role j . Summing a PRE path and using the pre-lag Taylor bound expresses its selected physical argument, up to the controlled uniform error, as

$$e(c_j p^{\rho_j}(y_j)) e(\gamma_j^*(y_j - x_*)) g_j(\xi(y_j - x_*)).$$

The function g_j is a cutoff times a trigonometric polynomial. After collecting the common multiplier (5.4), all Fourier indices in any anchor coordinates lie in the box of radius B'_h , enlarged by fixed factors depending on b . This is H(29) with its complete coefficient-sum and error bounds.

To compare physical and torus amplitudes, take the horizontal tiny space K_B of a nonexceptional primary in a successful component, and let

$$\mathbb{H} = (K_B \cap \mathbb{Z}^{\dim \theta_z})^\perp \subseteq \mathbb{T}^{\dim \theta_z}.$$

This is a connected torus because the integer lattice being annihilated is saturated. The tiny/huge alternative makes physical interval averages of the relevant characters agree with Haar averages on $\mathbb{H}$, to the prescribed small error. The component floors above absorb those errors, giving

H(30):

$$a'_j := \|g_j\|_{L^2(\mathbb{H})} \leq C_b \left[\sum_{\mathcal{D} \text{ selected}} \left(\|A_{\mathcal{D}}\|_{2,I} + D_*^{-30} \sum_{\substack{m \in \mathcal{D} \\ m \text{ core}}} |\alpha_m| \right)^2 \right]^{1/2} \leq C_b \sigma_{\mathcal{F}_j}.$$

The selected sets are whole minimal CURRENT subtotals, by coherence of the height predicates on $\mathcal{H}_{\text{up}}$. For distinct roles, $u + (b_j - b_{l'})v$ and $u + (b_i - b_{l'})v$ have independent Haar distributions on $\mathbb{H}^2$. Interpolating the six two- L^1 bounds gives

$$(5.11) \quad \int_{\mathbb{H}^2} \prod_j |g_j(u + (b_j - b_{l'})v)| \, du \, dv \leq \prod_j a'_j.$$

A bounded progression multiplier changes only the constant.

For four core roles put

$$Q = \sum_j \gamma_j^*, \quad T = \sum_j b_j \gamma_j^*.$$

Choose, by a fixed rule independent of the mode masks, a reference $q' = Q + n \cdot \xi$, $t' = T + m \cdot \xi$ in the full Fourier box with $\max(|q'|, |t'|) \leq U_0 R_s$, where $U_0 = D_*^{250}$. Introduce dyadic parameters by

$$(5.12) \quad \mu \asymp \min \left(1, \min_j \frac{|t' - b_j q'|}{R_s} \right), \quad \nu \asymp \max \left(1, \frac{|q'|}{R_s}, \frac{|t'|}{R_s} \right).$$

Discard absent references and inner minima below U_0^{-1} . The four marginals of the normalized kernel make its Fourier transform vanish on $t' = b_j q'$. The mean value theorem near these lines and rapid Fourier decay give $C_{L,b} \mu \nu^{-L}$ for every fixed L . The tiny/huge dichotomy puts all moderate frequencies in one pair of character cosets modulo K_B ; summing that pair of cosets is a Haar integral bounded by (5.11). Thus its contribution is at most $C_{L,b} \mu \nu^{-L} \prod_j a'_j$.

For three core roles with side role l , anchor at y_l and put

$$Q = \sum_{j \neq l} \gamma_j^*, \quad T_l = \sum_{j \neq l} (b_j - b_l) \gamma_j^*.$$

Choose a moderate reference $T_l + m \cdot \xi$ and write $\beta = r(T_l + m \cdot \xi)$. Set $\mu \asymp \min(1, |\beta|)$, $\nu \asymp \max(1, |\beta|)$, with the same small and large discards. Partial Fourier integration of the kernel in the step variable gives a function W_β whose fixed-order derivatives are bounded by $C_{L,b} \mu \nu^{-L}$. At zero this uses the l -marginal; at large frequency it uses integration by parts. Duality in (5.11) bounds the remaining horizontal coefficient sequence in ℓ^2 .

More explicitly, average a fresh offset cutoff ζ^{ρ_l}/c_ζ in the side, where $c_\zeta = \mathbb{E}_{\rho_l} \zeta^{\rho_l} \geq 1/2$ is constant. This insertion is an exact identity in expectation. Keep the bounded factor c_ζ^{-1} outside the estimate and set $S' = \zeta^{\rho_l} e(-c_l p^{\rho_l}) S_{\text{phys}}$ on I . Its normalized L^2 norm is at most that of the side. Write $\widehat{S}'_I(k')$ for its normalized Fourier coefficients. The horizontal pairings decay in $|k' + r(Q + n \cdot \xi)|$. Schur's estimate on a dyadic shell of radius Λ , followed by averaging the fresh offset, gives H(32) with

$$(5.13) \quad b_\Lambda^2 = \mathbb{E}_{\rho_l} \sum_{\substack{k' \in \mathbb{Z}: \\ |k' + r(Q + n \cdot \xi)| \leq 2\Lambda \text{ for some } n, \\ |n|_\infty \leq B'_h}} |\widehat{S}'_I(k')|^2.$$

The contribution is bounded by $C_{L,b} \mu \nu^{-L} \prod_{j \neq l} a'_j \sum_{1 \leq \Lambda \leq U_0} \Lambda^{-L} b_\Lambda$, where the sum is dyadic. Notice that the windows use the whole Fourier box and are independent of the mode masks.

The error estimates in [H, Section 9.1–9.2] now apply with these explicit tests. They use the absolute local form estimate, Fourier tails, the pre-lag, and the rational gap, all with reserves already chosen above. Per tuple the error is bounded by a fixed multiple of D_*^{-100} times the absolute core coefficient sums, and by the side size if a side is present. The common-height argument samples heights independently and uniformly from $\mathcal{H}_{\text{up}}$ until all roles are good; its first-success weights sum to one on each pure tuple. A core mode has bad fraction at most $2\delta_*$, so the sum over failure assignments converges. Consequently these errors sum absolutely to $O(e^{-k}|R|)$, and the principal cell terms have the form H(33):

$$(5.14) \quad C_{L,b}\mu\nu^{-L}\Lambda^{-L} \left(\prod_{j=0}^3 \sigma_j \right) N_s^G.$$

Here the σ_j are dyadic component-amplitude bins, with the side bin belonging to b_Λ ; N_s^G counts the representative tuples passing the tests. For four cores, $\Lambda = 1$. We next bound these counts across scales.

5.5. Sparse mass for separated tests. The finite counting statement is useful independently of the mode construction. Its integer divisibility is the reason for the choice (5.5).

Lemma 5.3 (Sparse mass). *Let $J \subseteq \{0, 1, 2, 3\}$ consist of three or four core roles. In each role let a finite label set $\mathcal{A}_i$ be identified injectively with real numbers, with $|\mathcal{A}_i| \leq M$, $M \geq 1$. Let $d \geq 0$ be an integer. Fix $\zeta \in \mathbb{R}^d$, $B, L \geq 1$, $Q, \Sigma, R_0 > 0$, and $0 < \mu \leq 1 \leq \nu$. Every shift vector below belongs to $\mathbb{Z}^d$. For finitely many distinct integers s , at dyadic scales $R_s = R_0 2^s$, let $\mathcal{E}_s \subseteq \prod_{i \in J} \mathcal{A}_i$ and $N_s = |\mathcal{E}_s|$. Assume the following five conditions.*

- (1) *Two labels in different core roles determine at most one full tuple, and that pair occurs at at most L scales.*
- (2) *If there is a side role, there are $w_s \geq 0$, $\sum_s w_s \leq M$, such that every fixed core label is incident to at most w_s tuples at scale s .*
- (3) *Distinct same-role labels occurring at scales s, t satisfy*

$$|a - a' + n \cdot \zeta| > \Sigma \max(R_s, R_t) \quad (|n|_\infty \leq Q).$$

- (4) *For every triple $T \subseteq J$, with excluded role $l' \in \{0, 1, 2, 3\} \setminus T$ and $\alpha_i = b_i - b_{l'}$, each tuple has a witness $|n|_\infty \leq B$ such that*

$$\mu R_s \leq \left| \sum_{i \in T} \alpha_i a_i + D_s^2 n \cdot \zeta \right| \leq \nu R_s.$$

- (5) *At every occupied scale, $|n \cdot \zeta| \notin [\mu R_s/1000, 1000\nu R_s]$ for all $|n|_\infty \leq Q$.*

There are constants $C_b, C_{\text{res},b}$ such that, if

$$Q \geq C_{\text{res},b}(d+1)^2 B, \quad \Sigma \geq C_{\text{res},b}(d+1)^2 \nu,$$

then, with $\ell = 1 + \log_2(2 + (d+1)^2 \nu/\mu)$,

$$\sum_{s: N_s \leq \delta M^2} N_s \leq C_b \ell L^{29} \delta^{1/2080} M^2 \quad (0 < \delta \leq 1).$$

The exponents and constants are independent of the dimension and the number of labels and scales, apart from the indicated factors.

Proof. We follow the finite argument of [H, Lemma 9.1], spelling out the arithmetic and the dimension dependence. The proof first covers most label differences by neighborhoods associated with later scales. Three such future covers then place sums of triples in a thin slab. Separated anchor labels reduce that slab's cover from three factors of M to two, while the pure-shift gap

makes many translated triples distinct inside the cover. This comparison forces small mass at sparsely occupied scales.

Approximate all finite real data on a sufficiently fine common grid, taking approximants on a fixed divisible multiple of its step so divisions by the α_i remain on the grid. For example, a multiple of D_s^2 suffices. Choosing the grid after the data preserves separation and annular inequalities with fixed relaxed margins and leaves a narrower pure-shift gap. Round scalar widths upward by at most one grid step. The estimates below are uniform in that step.

Restrict to $N_s \leq \delta M^2$. Pair determination gives total mass at most LM^2 . For parameters $\delta \leq \alpha/2$ and $0 < \kappa < 1$, greedily partition the scales in decreasing order of R_s into counting blocks of mass at most αM^2 . There are at most $J_0 = 3L/\alpha$ blocks. For a block $\mathfrak{b}$, let $R_{\mathfrak{b}}^+$ be its largest scale, take $N_0 = \lceil C_b B \rceil$ with $C_b > 32D_s$, and choose a grid width $\epsilon_b \asymp \nu R_{\mathfrak{b}}^+$. Use the finite neighborhoods

$$U_{\mathfrak{b}}(t) = \{D_s n \cdot \zeta + u : |n|_{\infty} \leq tN_0, u \text{ on the grid, } |u| \leq t\epsilon_b\}.$$

They are symmetric and satisfy $U(t) + U(t') \subseteq U(t + t')$.

Fix a role and a counting block with uncovered mass exceeding κM^2 . Project onto this role and another core role, retaining one scale witness per pair, and color edges by a third core label. The graph has at least $(\kappa/L)M^2$ edges and a proper coloring with at most M colors. The elementary four-step-walk observation in the proof of [H, Lemma 9.1] gives a subset B_0 of at least $c\eta M$ vertices, $\eta = \kappa/L$, such that any two have at least $c\eta^5 M^3$ four-step walks with distinct ordered color tuples. Repeated vertices are allowed. That observation is purely a graph statement: choose a random neighborhood, remove vertices with many low-codegree partners, and count the two choices of a common neighbor on either side of a retained midpoint.

Alternating the four annular relations along a walk cancels the intermediate labels. Dividing by the nonzero integer α_i gives

$$a - a' \in q(c_1, c_2, c_3, c_4) + U_{\mathfrak{b}}(1).$$

Here q depends only on the ordered four edge colors; these c_i are graph colors, unrelated to the fixed quadratic weights. The scalar error after division is at most $4\nu R_{\mathfrak{b}}^+$, and D_s^2/α_i is an integer multiple of D_s , with its index covered by N_0 . Packed differences outside $U_{\mathfrak{b}}(2)$ cannot use the same color tuple. The M^4 possible color tuples therefore bound their number by $C\eta^{-5}M$. Repeating after removing B_0 covers all but κM^2 occurrences in each role and block by at most $K_1 = C(L/\kappa)^6$ subsets, each with its difference set covered by $K_1 M$ translates of $U_{\mathfrak{b}}(2)$.

Discard uncovered occurrences. Call a core label future when it belongs to a cover subset in a later counting block. Nonfuture label sets are disjoint over blocks. Let $m_i^{\mathfrak{b}}$ be their cardinalities in core role i . Treat the side as nonfuture, with $m_i^{\mathfrak{b}} = \sum_{s \in \mathfrak{b}} w_s$. Occurrences with fewer than three future roles have two nonfuture roles. Their total mass is at most $C\sqrt{L}\alpha M^2$, by summing $\min(\alpha M^2, Lm_i^{\mathfrak{b}}m_j^{\mathfrak{b}})$ and Cauchy-Schwarz. The discarded uncovered mass is at most $4J_0\kappa M^2$.

Assign each remaining occurrence three future roles and one future cover subset in each. A class fixes its current counting block and its three future block/subset choices. There are at most $4J_0^4 K_1^3$ classes. Fix one of mass ρM^2 , with future subsets B_1, B_2, B_3 . The subscripts now enumerate those roles; their coefficients remain the corresponding α_i . Set $\Gamma = C_{\text{gap},b}(d+1)^2\nu/\mu$. If $\rho \leq C_b\delta\ell$, retain this bound. Otherwise split the mass into four consecutive ranges by quartiles, omitting the at most three scale bins crossing a cut. Each range retains at least $\rho M^2/8$ mass. The first provides anchor sets $B_{i0} \subset B_i$ of size at least $c\rho M/L$. Let $R^{\#}$ be the largest scale of the second range; these anchors are separated modulo reserved shifts at scale $R^{\#}$. The fourth range gives a set F of at least $c\rho M^2/L$ distinct triples projected onto these three roles. The third range supplies the scale margin, and a greedy thinning of the second gives $q \geq c_b\rho/(\delta\ell)$ triples

t_v , also projected onto these roles, at scales separated by factors at least Γ . All future-block scales and fourth-range scales are smaller than these by at least the required factor Γ .

Write U_i for the neighborhoods of the three future subsets. Subdividing index and scalar intervals covers $U_i(t)$ by at most $(2t+1)^{d+1}$ translates of $U_i(1)$. Hence there is an odd $H_i \leq 100(d+1)^2$ with $|U_i(H_i+2)| \leq 2|U_i(H_i)|$. The preliminary exponential cover is used only to find this small-growth index, and is never paid in the cardinality estimate. The translates $a + U_i(H_i)$, $a \in B_{i0}$, are disjoint: an intersection would contradict the anchor separation with a shift index $O_b((d+1)^2B)$ and scalar error $O_b((d+1)^2\nu R^\#)$.

Set $Z_i = B_{i0} + U_i(H_i)$. The difference covers give $|Z_i - B_i| \leq C(K_1L/\rho)|Z_i|$. Choose a nonempty $X_i \subset Z_i$ minimizing $r_i = |X_i - B_i|/|X_i|$. Petridis's minimal-growth subset argument, in the form reproduced in [H, Lemma 9.1], gives $|X_i - mB_i| \leq r_i^m|X_i|$. One anchor translate contains a subset $Y_i \subseteq X_i$ of at least $|X_i|/M$ points. If $\mathcal{P} \subseteq mB_i$ has all distinct differences outside $U_i(2H_i)$, the sets $Y_i - p$, $p \in \mathcal{P}$, are disjoint and lie in $X_i - mB_i$. Thus $|\mathcal{P}| \leq r_i^m M$. A maximal packing covers mB_i by that many translates of $U_i(2H_i)$. Only $m = 2, 3$ is used.

The outputs $t_v + F$ lie in the slab

$$(x_1, x_2, x_3) \in 2B_1 \times 2B_2 \times 2B_3, \quad \left| \sum_i \alpha_i x_i + D_s^2 n \cdot \zeta \right| \leq 2\nu R^\#, \quad |n|_\infty \leq 2B.$$

Put $r_0 = CK_1L/\rho$. Cover the first two sumsets by $r_0^2 M$ boxes each. Above a fixed pair, dividing the slab equation by α_3 places the third coordinate in

$$c + \{n \cdot \zeta : |n|_\infty \leq C_b(d+1)^2 B\} + [-C_b(d+1)^2 \nu R^\#, C_b(d+1)^2 \nu R^\#].$$

Every shift remains integral: both D_s^2/α_3 and $D_s\alpha_i/\alpha_3$ are integers. If $\mathcal{P}$ is a packing of the local third-coordinate set outside $U_3(2H_3)$, then $B_{30} + \mathcal{P} \subset 3B_3$ is still a packing. Collisions with different anchors are excluded by the same shift and scalar bounds. Thus $|B_{30}||\mathcal{P}| \leq r_0^3 M$. The entire slab is covered by at most $CK_1^7(L/\rho)^8 M^2$ product boxes.

Each such box contains at most one of the indexed outputs $t_v + F$. Within one translation this follows from label separation. Across translations, let R_{big} be the larger selected scale. Its annular residual has magnitude at least μR_{big} ; the other three residuals together are at most $\mu R_{\text{big}}/10$. Membership in one box adds scalar error at most $C_b(d+1)^2 \nu \max_i R_{\text{future},i}^+ \leq \mu R_{\text{big}}/10$, by the choice of $C_{\text{gap},b}$. The resulting pure shift has magnitude between $4\mu R_{\text{big}}/5$ and $3\nu R_{\text{big}}$, with index $O_b((d+1)^2B) \leq Q$, contradicting the forbidden gap.

Comparing $q|F|$ with the slab cover yields $\rho^{10} \leq C_b \delta \ell K_1^7 L^9$, and hence $\rho \leq C_b \ell K_1 L \delta^{1/10}$. Including all choices and losses gives

$$M^{-2} \sum_{s: N_s \leq \delta M^2} N_s \leq C_b \left(\frac{L\kappa}{\alpha} + \sqrt{L\alpha} + \ell L^{29} \alpha^{-4} \kappa^{-24} \delta^{1/10} \right).$$

Take $\alpha = \delta^{1/1040}$, $\kappa = \alpha^2$ for sufficiently small δ , and use the crude mass bound for the remaining values. This proves the stated exponent $1/2080$. $\square$

5.6. Application of the count and the resulting core bound. We verify the hypotheses rather than apply the original arithmetic form of [H, Proposition 9.2]. Fix a path, height, role types, frequency bands, and amplitude bins in (5.14). Identify each representative label with its value $\omega_a(x_*)$, which is injective in a fixed role and type by (5.8). Define

$$M_i^G = |\mathcal{A}_i^G| \quad \text{for core roles,} \quad M_l^G = \sigma_l^{-2} \sum_s \|S_{l,G,s}\|_{2,I_s}^2 \quad \text{for a side,} \quad M' = \max_i M_i^G.$$

Groups with a nonzero count have $M' \geq 1$.

Two fixed core labels determine the others. For four cores, subtract the equations for Q, T of two candidate tuples. The two unknown label differences solve a system with determinant $b_j - b_i$, modulo bounded rational shifts and errors $O_b(\nu \max(R_s, R_t))$. For three cores the equation for T_l determines the last label, dividing by $b_j - b_l$. The shift reserve and separation force these differences to vanish. For every core triple with excluded role l' , the tests also give the annulus

$$(5.15) \quad c_b \mu R_s \leq \left| \sum_i (b_i - b_{l'}) \omega_{a_i}(x_*) + n \cdot \xi \right| \leq C_b \nu R_s, \quad |n|_\infty \leq C_b B'_h.$$

For four cores this is $t' - b_{l'} q'$; for three it is the T_l test. Once the labels are fixed, the height-bin lemma [H, Lemma 3.5] bounds their scale occurrences by $L_0 \leq P^C (1 + \log(\nu/\mu))^C$.

If one role is a side, windows (5.13) for different triples incident to a fixed core label are disjoint. An overlap gives both a relation for their Q values with error $O(\Lambda R_s)$ and their two T_l relations. Solving the resulting two-by-two system again divides by a $b_j - b_i$; separation then identifies the tuples. Couple the fresh offset across these windows. Parseval shows their incidence count is at most $w_s = \sigma_l^{-2} \|S_{l,G,s}\|_{2,I_s}^2$, giving the side-budget hypothesis.

Here are the promised reserve checks for all these divisions. In Lemma 5.3 take

$$\zeta = \xi/D_s^2 = D_s^{\Delta-2} \vartheta/L_h, \quad Q_{\text{res}} = \left\lfloor E_h/D_s^{\Delta-2} \right\rfloor, \quad \Sigma = \Xi/2.$$

Every divided ξ -shift is integral in this finer lattice, because each determinant divides D_s . The fixed-power choice of A_0 ensures

$$Q_{\text{res}} \geq C_{\text{res},b} (d+1)^2 C_b B'_h, \quad d = \dim \theta_z.$$

Also $\nu, \Lambda \leq C_b D_*^{250}$, while $\Xi = D_*^{1000}$, so the separation reserve holds. All rational shifts remain below $\mathcal{H}$. Every successful component contains a nonexceptional primary, even if its representative is exceptional. Applying (5.7) to its pure horizontal shifts excludes the entire intermediate interval required by the pure-shift hypothesis, since $\mu \geq U_0^{-1}$ and the interval length $|I_s|$ lies between $r_{t_{s_0}}$ and $r_{v_{s_0}}$. This checks all five hypotheses.

Lemma 5.3, pair determination, and the side incidence bound therefore give the two counts H(34): for some fixed $p_0 < 1$,

$$(5.16) \quad \sum_s N_s^G \leq L_0 \min_{i \neq j} M_i^G M_j^G,$$

$$(5.17) \quad \sum_s (N_s^G)^{p_0} \leq P^C (1 + \nu + \mu^{-1} + \Lambda)^C (M')^{2p_0}.$$

Indeed the sparse-mass exponent $c' = 1/2080$ allows any $1 - p_0 < c'$. On the bins $2^{-j-1} (M')^2 < N_s^G \leq 2^{-j} (M')^2$, apply the sparse bound and sum the geometric series with ratio $2^{1-p_0-c'} < 1$.

Finally the amplitude and group summation of [H, Proposition 9.3] uses precisely (5.14), (5.9), and (5.16)–(5.17). Interpolate the two counts with a sufficiently small fixed weight $\theta' > 0$, giving power $p = (1 - \theta') + \theta' p_0$. The path square budgets sum the group factors, and the core count cap D_* controls the common amplitude shift. The three amplitude-gap sums are geometric, as in that proposition. The frequency losses after interpolation are at most $\mu^{-C\theta'} \nu^{C\theta'} \Lambda^{C\theta'}$; choose θ' small and then the derivative order L large so they are summable against $\mu^p \nu^{-Lp} \Lambda^{-Lp}$. Applying the original cell minimum before subadditivity yields

$$(5.18) \quad C_{q,b} \eta^\varepsilon P^{C_4} |R| + O(e^{-k} |R|)$$

for the terms with at least three core arguments, with fixed $\varepsilon > 0$ and C_4 independent of q . The same calculation is the core step of [V, Section 6.6]: its frozen scheduled inputs provide exactly the amplitudes, representatives, and same-cell side incidence verified above. Together with

Proposition 5.1 and the side square bound, this supplies the side and core estimates required to close the proof of Theorem 2.2.

6. PARAMETER CHOICE AND COMPLETION OF THE LOCAL ESTIMATE

The preceding sections provide the coefficient-dependent inputs to the completion arguments in [H, Section 10] and [V, Section 7]. We now explain the common parameter choice and the two distinct conclusions obtained from it. First we bound the fixed-input constant $C_N(q)$ independently of N . Only after this absorption do we complete the proof for depth-dependent inputs.

6.1. The retained-cell estimate. We use the base reductions of [H, Proposition 5.8] and [V, Proposition 5.6], with the modifications proved above. At one noninitial accuracy level their target, on a normalized root R , is a sum of terms $|I| \min\{\eta, |H_I(z_I)|\}$ over a specified subset of descendant cells. Here z_I is the four-tuple supplied by the base expansion at I , and $\eta = \exp(-k^\theta)$ is its quality parameter, with k the accuracy log and $0 < \theta < 1$ fixed. The constructions then decompose each of these four-tuples into the localized terms estimated in the preceding sections.

The stopping constructions in [H, Section 10.1] and [V, Section 7.1] are used in their stated order. Main columns and their uses are fixed first; mode expansions and the whole side functions are fixed next; helper projections are constructed after those side functions. Size and counting stops are then imposed on these fixed objects. A stop removes original summands from an attempt; it does not redefine a previously fixed stack function. In particular, the maximal functions in the short-block estimates remain defined on whole retained cells, including the portions below later stopping cells.

Let $\mathcal{A}_R^{\text{ret}}$ denote the cells not removed by these stops in the attempt rooted at R . The conclusions of the preceding sections give the same estimate as the retained-cell calculations in the two companions:

$$(6.1) \quad \sum_{I \in \mathcal{A}_R^{\text{ret}}} |I| \min\{\eta, |H_I(z_I)|\} \leq \left[C_q \eta^\varepsilon P^{C_4} + (1 + C_N(q)) C_q P^{-1/2} G_0 (K + 2)^{C_{19}(q-2)} \right] |R| + O(e^{-k} |R|).$$

Here $\varepsilon > 0$ is fixed, P is the transfer parameter, K is the depth-block length, and G_0 is the row and stack overhead. The constants may depend on b, δ_0, C_0 .

For clarity, the estimate has three sources. Terms with at least three core factors use the powered core estimate, and terms with at least three side factors use their square-energy bounds. With $p = 1 - \varepsilon > 2/3$, both are converted to the first term of (6.1) by $\min\{\eta, a\} \leq \eta^{1-p} a^p$. Terms with exactly two sides use the transfer estimate for the whole complementary arguments. Its main term is again included in the first term. The remaining coupled near-band sum is estimated at power one by the short-block comparison; it gives the second term, including its factor $1 + C_N(q)$, by (5.10). Physical and numerical errors are estimated at power one as well. These alternatives exhaust the four-position expansion. The minimum is applied to the whole contribution of a cell before subadditivity is used; no smallness of a separately selected group is being assumed.

6.2. Restarts and the complexity hierarchy. The integrated size, path-count, and square-energy bounds allow the stopping thresholds to be enlarged by fixed factors so that the first stopped cells occupy at most $|R|/4$. At each such cell send the original summands there and below to a new attempt. The local normalizations and column bounds are available on every

new root. Linear-side energy is proved anew by the construction on that root, not by asserting a proportional restriction of a global energy bound. The resulting attempt roots satisfy

$$(6.2) \quad \sum_{R \text{ an attempt root below } R_0} |R| \leq \sum_{m=0}^{\infty} 4^{-m} |R_0| = \frac{4}{3} |R_0|.$$

All the trees are finite, so the restart argument can equivalently be read as induction on their remaining depths.

In the scheduled-input construction, the original depths, functions, and block partitions remain attached to the summands. Restriction to a new root preserves the pointwise size and variation bounds with the same W_j . The row estimates of [V, Section 4] therefore apply at each restart with no factor depending on W_j .

The distinction between the two complexity scales is essential. The bounds supplied by the constructions are

$$(6.3) \quad \begin{aligned} \log D_m &\leq C_q k^C, & \log D_* &\leq C_q (1 + k + P)^C, \\ K &\leq C_q (1 + k + P)^{C_5}, & G_0 &\leq \exp(C_q (1 + \log(2k))^C). \end{aligned}$$

Here D_m bounds the main and helper catalogs, and D_* bounds the larger mode, Fourier, and rational-height data. The latter data enter the counting and mode estimates, but are not fed back into the main catalog construction. Strong stack estimates compress main projection families only; mode projections have already been included in the Bessel rows. This is precisely the separation in [H, Section 10.3] and [V, Section 7.3].

The coefficient modifications preserve this separation. Chart substitutions multiply heights and derivative bounds by fixed slope-dependent factors; the change of height base occurs only in the mode construction. The sparse-count estimate retains its dimension-polynomial reserves and its power saving. Thus all the degrees C, C_4, C_5, C_{19} can be fixed *before* choosing q . Their numerical values may be enlarged to cover both the fixed-input and scheduled-input constructions. Only coefficients such as C_q are allowed to depend on the later choice of q .

Choose $q \in (2, 3)$ sufficiently close to 2 that

$$(6.4) \quad C_{19} C_5 (q - 2) < \frac{1}{8}$$

for both constructions. This common choice is legitimate because local detection is available for every $q > 2$ and short-block comparison for every $2 < q < 3$. Next put $P = \lceil \eta^{-\alpha} \rceil$, choosing a fixed $\alpha > 0$ so small that $\alpha C_4 < \varepsilon/2$, as well as the analogous finitely many upper bounds already included in the powered costs. Finally choose the starting accuracy sufficiently large. Since P dominates every fixed power of k and $G_0 = P^{o(1)}$, we obtain

$$P^{-1/2} G_0 (K + 2)^{C_{19}(q-2)} \leq C_q P^{-1/4}, \quad \eta^\varepsilon P^{C_4} \leq C_q \eta^{\varepsilon/2}.$$

Also $P^{-1/4} \leq \eta^{\alpha/4}$, and e^{-k} is smaller than these quality powers for large k . Consequently, for some $c_* > 0$, (6.1) and (6.2) give the full base-segment bound

$$(6.5) \quad \sum_{I \text{ in the base segment}} |I| \min\{\eta, |H_I(z_I)|\} \leq C_q \eta^{c_*} (1 + C_N(q)) |R_0|.$$

6.3. Fixed inputs: accuracy summation and absorption. The base accuracy levels satisfy $k_{t+1} = k_t^A$ for a fixed $A > 1$. For tuples whose later level is t , there are $O(t)$ choices of the earlier level and a bounded number of slot patterns. The segment-root packing in [H, Section 5], followed by (6.2), gives a bounded total root length for each tuple. The noninitial terms in (6.5)

therefore have total coefficient controlled by

$$(6.6) \quad C_q \sum_{t \geq 12} t \exp(-c_* k_t^\theta).$$

This series is finite, and its value is arbitrarily small when the starting accuracy is sufficiently large. The replacement and two-increment errors have the same summable multiplicity. The finitely many initial levels have depth-independent bounds from the active-scale reduction, as in [H, Section 10.4]. Thus the modified construction proves the following normalized conclusion.

Proposition 6.1 (Normalized absorption). *For the fixed q chosen above and every sufficiently small $\tau > 0$, there is $A_\tau < \infty$, independent of N , such that the following holds. Suppose a finite rooted dyadic subtree has root I_* , its summation cells lie in at most N consecutive depths, and its fixed scalar inputs satisfy*

$$\left(|I|^{-1} \int_I |f_j|^q \right)^{1/q} \leq 1 \quad (0 \leq j \leq 3, I \text{ in the subtree}).$$

Then, for any subset of summation cells and admissible kernels,

$$\sum_{I \text{ a summation cell}} |I| |H_I(f_0, f_1, f_2, f_3)| \leq (A_\tau + \tau C_N(q)) |I_*|.$$

The argument just given is the proof: take the coefficient in (6.6) below τ and include its constant part, the initial levels, and the numerical errors in A_τ . The base expansion is first taken at a finite accuracy cutoff. At each fixed cell the terminal residual tends to zero by the pair-comparison estimates of the base reduction; since there are only finitely many cells, the passage to all accuracy levels is legitimate. This is the concluding expansion step in [H, Section 10.4].

We include the outer normalization argument to make the absorption explicit. Start with disjoint roots at the largest scale in a finite collection from (2.7). At a root J , write $a_{j,J} = (|J|^{-1} \int_J |f_j|^q)^{1/q}$. If some $a_{j,J}$ vanishes, all forms below J vanish. Otherwise stop at the first strict descendants J' on which an average of $|f_j|^q$ exceeds $16a_{j,J}^q$. The union of first stops has measure at most $|J|/4$. Dividing f_j by $16^{1/q} a_{j,J}$ normalizes the intervening tree, so Proposition 6.1 bounds its sum by

$$16^{4/q} (A_\tau + \tau C_N(q)) |J| \prod_j a_{j,J}.$$

Repeat at the stopped roots. The sets E_J left after deleting their first stops are disjoint, have measure at least $3|J|/4$, and satisfy $M_q f_j \geq a_{j,J}$ on E_J . Hence

$$\sum_J |J| \prod_j a_{j,J} \leq \frac{4}{3} \int_{\mathbb{R}} \prod_j M_q f_j.$$

Taking the supremum in the definition of $C_N(q)$ gives

$$C_N(q) \leq C A_\tau + C \tau C_N(q),$$

where C is independent of N and bounded for $2 < q < 3$. Choose $C\tau < 1/2$ and use the already established finiteness of $C_N(q)$ to absorb the last term. We have proved

$$(6.7) \quad C_H(q) := \sup_{N \geq 1} C_N(q) < \infty.$$

6.4. Depth-dependent inputs. Now perform the scheduled-input construction with the same q , using its own subsequent choices of α and starting accuracy. Insert (6.7) into (6.5). The factor $1 + C_N(q)$ is now a fixed constant. The scheduled base expansion of [V, Section 5] has the same accuracy multiplicity and segment-root packing, with the depth-row estimates supplying the uniform bounds. Its noninitial terms therefore sum by (6.6). Its finitely many initial levels are bounded by its active-scale reduction independently of the number of depths and of every W_j .

Finally, at a fixed original cell, all accuracy updates are evaluated on its same actual scheduled input $z_{j,s(I)}$. The per-cell pair comparisons therefore make the terminal residual tend to zero. There are finitely many cells, so the finite-cell sum commutes with this limit. The triangle inequality and the uniform summed bounds give (2.4), exactly as in [V, Section 7.4]. This proves Theorem 2.2, and hence, by Section 3, Theorem 1.1.

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